

# NCA Assistant

## User Manual

Version 1.9

Non-Compartmental Pharmacokinetic Analysis

Bioequivalence • Drug Interaction Studies • Power & Sample Size

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Not validated for your use: qualifying your installation is your responsibility (see Intended Use and Your Data).

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## Intended Use and Your Data

NCA Assistant is for pharmacokineticists doing non-compartmental analysis, average bioequivalence testing and study planning. It gives no reference-scaled bioequivalence verdict (ABEL, RSABE). The public instance and a standard installation have no audit trail, electronic signature or access control; a controlled installation on your organisation's server adds them (Working on a Controlled Installation). The public instance on shinyapps.io is for evaluation and training, with synthetic or pseudonymised data. For regulated work, install a tagged release on your own system and qualify it there with the validation package. Responsibility for the analysis and its conclusions stays with the user. Not for dosing decisions for individual patients.

**Your data.** On the public instance, uploads are processed on shinyapps.io servers run by Posit PBC (USA). Upload only synthetic, example or anonymised data there. Pseudonymised trial data are still personal data under the GDPR. Sending them to a third-party host needs agreements your organisation must have in place, and may breach sponsor confidentiality. For real study data, run the app on your own computer.

**Status.** NCA Assistant is not validated for your use. The validation package in the repository (a URS, an IQ/OQ/PQ protocol and a test script with more than 400 tests) is a starting point: the developer's test run shows that the code does what the tests check, on the developer's computer. Qualifying the installation you use is your responsibility; the chapter on regulated use explains how.

**Supplier, support and warranty.** The app is free, open-source software under the GNU General Public License, version 3 or later, written by Rob ter Heine (Radboud Applied Pharmacometrics, Radboudumc, Nijmegen). It comes without any warranty and without a support agreement. Problems and questions go to the issue tracker at [github.com/Robterheine/NCAassistant/issues](https://github.com/Robterheine/NCAassistant/issues).

## What the App Does Not Do

Use other software for the following. The app refuses some of these, but not all, so check before you start.

| Not supported                                                                                                                   | What to use or do instead                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference-scaled bioequivalence verdicts (EMA ABEL, FDA RSABE) and the FDA NTID scaled test                                     | Software that implements them, for example the replicateBE package for ABEL. The app shows CVwR and the limits it would imply for information only.                               |
| Studies with three or more treatments in one analysis (for example a three-way food-effect or DDI crossover, a Williams design) | Analyse two treatments at a time (see Studies with Three or More Treatments), knowing the residual then comes only from those periods, or use software that fits the full design. |
| Dose-proportionality statistics (the power model)                                                                               | Use the dose-normalised parameters and plots here, and fit the power model elsewhere.                                                                                             |
| Absolute or relative bioavailability between routes or strengths in one comparison                                              | Run the NCA per route with dose normalisation and compute F outside the app.                                                                                                      |
| Urine data (amount excreted, renal clearance, excretion rate)                                                                   | Other NCA software.                                                                                                                                                               |

|                                                                                                                                                       |                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Sparse or destructive sampling (one sample per subject, Bailer's method)                                                                              | Other NCA software, or a population approach. |
| Therapeutic drug monitoring data with irregular dosing, superposition, several infusion durations in one run, baseline-corrected endogenous compounds | A population PK model, or other NCA software. |
| Dosing decisions for individual patients                                                                                                              | Not an intended use.                          |

## Getting Started

### Opening the Public App

Go to [robterheine.shinyapps.io/NCAassistant](https://robterheine.shinyapps.io/NCAassistant) in Chrome, Firefox, Edge or Safari. There is nothing to install. The public instance is for evaluation and training: read Intended Use and Your Data before you upload anything. It disconnects after a period without activity, and it keeps nothing between sessions.

### Installing the App on Your Own Computer

For confidential data and for regulated work, run the app on your own computer. It needs R 4.1 or newer; RStudio is optional but makes the steps easier.

1. Install R from [cran.r-project.org](https://cran.r-project.org), and optionally RStudio Desktop from [posit.co](https://posit.co).
2. Download a release from [github.com/Robterheine/NCAassistant/releases](https://github.com/Robterheine/NCAassistant/releases): under Assets, choose "Source code (zip)", and unzip it. Use a tagged release rather than the latest code, so you know which version you run.
3. In RStudio, open `install_and_run.R` from the unzipped folder and click Source. Without RStudio, open a terminal in that folder and run `Rscript install_and_run.R`.
4. The first time, the script installs about twenty R packages from CRAN, which can take several minutes. Later starts take a few seconds.
5. The app opens in your web browser. To stop it, press Esc in the RStudio console, or Ctrl+C in the terminal.

For a qualified installation, add `--validated` (`Rscript install_and_run.R --validated`), or run `NCA_VALIDATED <- TRUE` before sourcing the script. It then installs the package versions the release was validated with, listed in `validation/renv.lock`, instead of the latest ones from CRAN. Run it from the folder that contains `app.R`.

**If the installation fails.** Read the last lines the script printed: they name the package that failed. On Windows, a package that needs compiling may need Rtools ([cran.r-project.org/bin/windows/Rtools](https://cran.r-project.org/bin/windows/Rtools)); on a Mac, the Xcode command-line tools (`xcode-select --install` in a terminal). Then run the script again; packages that installed are not installed twice.

### The Hub Screen

The hub has six cards, numbered in the order of a typical study. Plan a Study needs no data. Upload & Check Data is where every analysis starts; the other four paths use the data processed there.

| Card                   | Needs uploaded data | What you get                                             |
|------------------------|---------------------|----------------------------------------------------------|
| 1. Plan a Study        | No                  | Sample size or power, with plots                         |
| 2. Upload & Check Data | (starts here)       | A quality report and the processed data for paths 3 to 6 |
| 3. Visualize Data      | Yes                 | Concentration-time figures and a Figure Record           |

|                          |                          |                                                                            |
|--------------------------|--------------------------|----------------------------------------------------------------------------|
| 4. One Subject at a Time | No (manual entry) or yes | NCA of one profile, with the half-life inspector                           |
| 5. All Subjects (Batch)  | Yes                      | NCA of every profile, summary statistics, a Complete Analysis Record       |
| 6. Bioequivalence        | Yes                      | NCA, ANOVA, confidence intervals, a verdict and a Complete Analysis Record |

Once data are processed, the navigation bar shows a badge such as "12 subjects, 132 obs". Home, in the bar under the navigation, brings you back to the hub. Processing a new file clears every result of the previous one, and a result is also cleared when you change a setting it depends on: run the analysis again after a change. Settings stay when you leave a page and come back.

## Getting Help Inside the App

Every setting that needs explaining has an ⓘ button next to it; click it for a short explanation in plain language. The top navigation bar has three more sources. The Data Guide shows how to lay out the data for each study type, with example files to download. Statistical Methods describes every calculation, with references, and ends with Example Methods Paragraphs to adapt for a report or thesis. About lists the packages, their versions and licences, and the version history.

## Saving Your Work

The app keeps nothing between sessions. Before you close the browser tab, or leave the public app idle, download what you need: the results as Excel or CSV, the figure, or the Complete Analysis Record, which holds the data, the settings and a script that reproduces the analysis. After a disconnect, upload the file again and click Process Data. On a controlled installation, your records and the audit trail are kept on the server; the uploaded file and the results are not.

## Tutorials

Eight short tutorials, from a first analysis without a file to a bioequivalence study with partial AUCs, and one on reviewing a record on a controlled installation (Tutorial 8, in the chapter Working on a Controlled Installation). Each gives the steps with the labels the app shows, then questions. The answers are in the Answer Key at the end of this chapter, rounded as the app displays them. The example files are in the Data Guide (top navigation bar), under Example Datasets. The first time, do the tutorials in order: later ones assume you know the upload steps. Do the tutorials on the public app, on your own computer or on a training installation, never on a controlled production server, where every practice run would be recorded.

### Tutorial 1: A First Analysis Without a File

1. On the hub, click 4. One Subject at a Time.
2. Under **Where is your data?**, choose **Enter data manually (type values directly)**. The Time points and Concentrations boxes already hold an example profile: eight samples from 0 to 24 h, one value per line.
3. Click **Use This Data**.
4. Under **Analysis Settings**, keep Dose 100, Dose unit mg, Time unit h, Conc unit ng/mL and **Route of administration Oral / IM / SC (extravascular)**. Click **Run PK Analysis**.
5. Read the **PK Parameters** card, then the **Half-Life Calculation** card, which shows the points used for the terminal phase.

Questions: 1) What are C<sub>max</sub> and T<sub>max</sub>? 2) What is the AUC to last point? 3) What is the half-life, and on how many points does it rest? 4) What is the apparent clearance CL/F, and in which unit?

### Tutorial 2: Uploading Data with BLQ Results

This tutorial uses example\_blq.csv: eight subjects after a 100 mg oral dose, with results below the limit of quantification written as "<0.5".

1. On the hub, click 2. Upload & Check Data. Keep **What kind of file?** at **Simple table (one row per sample)**, and choose example\_blq.csv.
2. Check the column mapping: Subject ID is Subject, Time is Time, Concentration is Concentration.
3. Leave **LLOQ value** at 0 and click **Process Data**. The app stops: it found BLQ text but no LLOQ. It shows "LLOQ auto-detected: 0.5" with a button **Set LLOQ to 0.5**.
4. Click **Set LLOQ to 0.5**. The button only fills in the field; check that it reads 0.5, keep **BLQ handling rule** at the default (Rule 1), and click **Process Data** again. The quality report now lists the 22 BLQ entries as a warning.
5. Go to 5. All Subjects (Batch). Keep **Dose information** at **Same dose for all subjects**, enter 100 as **Dose given to each subject**, and check the units (mg, h, ng/mL).
6. Under **Partial AUCs (optional)**, set **Number of intervals** to 1, with Start 0 and End 1. Click **Run Analysis on All Subjects**.

Questions: 1) Which subjects have a lag time, and how long is it (Lag Time, in **All Parameters (per profile)**)? 2) Subject 5 has a BLQ result at 8 h between two measurable ones. What is its time of last measurable concentration? 3) What is subject 2's Partial AUC 0–1? Now go back to Upload & Check Data, choose Rule 6 (Before first quantifiable → LLOQ/2, rest → zero), click **Process Data**, and run the batch analysis again. What is subject 2's Partial AUC 0–1 now, and why did it change while the lag time did not?

### Tutorial 3: All Subjects at Once (Theophylline)

1. Download example\_theoph.csv and upload it. Map Subject ID to Subject, Time to Time and Concentration to conc. The app also maps the Dose column, which holds mg/kg. Leave **LLOQ value** at 0 and click **Process Data**; the badge shows 12 subjects, 132 obs.

2. Go to 5. All Subjects (Batch). Because a Dose column is mapped, **Dose information** is set to **Doses differ by subject or period (from Dose column in data)**. Switch it to **Same dose for all subjects** and enter 320 as **Dose given to each subject**, with Dose unit mg, Conc unit mg/L and Time unit h.

3. Keep **Route of administration** at **Oral / IM / SC (extravascular)** and **Trapezoidal method** at **Linear-up / Log-down**. Click **Run Analysis on All Subjects**.

Questions, with where to look: 1) Mean Cmax (**Summary Statistics**, Arithmetic Mean)? 2) Geometric mean AUC to last point (Summary Statistics, Geometric Mean)? 3) Median Tmax? 4) Which subject has the longest half-life, and how long is it (**All Parameters (per profile)**, sort by Half-Life)? 5) Geometric CV of AUC to infinity (observed) (Summary Statistics, Geometric CV (%))? 6) Is that longest half-life reliable? Tick **Show all parameters** and look at Points Used for Half-Life and AUC % Extrapolated (observed). 7) The note above the results names a subject with a pre-dose concentration above 5% of Cmax. Which subjects have a pre-dose concentration above zero, and what does that mean?

### Tutorial 4: Bioequivalence

1. Download example\_be\_crossover.csv and upload it. Map Subject ID, Time and Concentration, and also Treatment / Formulation to Treatment, Period to Period and Sequence to Sequence. Click **Process Data**.

2. Go to 6. Bioequivalence. In **Step 1: PK Analysis Settings**, enter Dose 100 with units mg, h and ng/mL, and keep the route at **Oral / IM / SC (extravascular)**.

3. In **Step 2: Bioequivalence Settings**, check that **Study design** is **2×2 standard crossover (TR | RT)**, that **Reference treatment** is Reference, and that **Statistical model** is **Fixed effects (all terms fixed)**. Keep **Log-transform (recommended)**, the **Parameters to compare** (Cmax and AUC to last point), **Confidence interval** 90 and the limits 80 and 125.

4. Click **Run Complete BE Analysis**.

Questions: 1) What are the ratio and 90% CI for Cmax (tab **Test vs. Reference Comparison**, columns Estimate and 90% CI Lower and Upper)? 2) And for AUC to last point? 3) Is the product bioequivalent for both, and what does the note above the table say? 4) What is the within-subject CV for Cmax? It is not in the table on screen: click **Download CI Table (CSV)** and take the Residual Variance (MSE), then  $CV = 100 \cdot \sqrt{(\exp(MSE) - 1)}$ ; or open Plan a Study, where a button offers the CV of this analysis.

## Tutorial 5: Planning a Study

1. On the hub, click 1. Plan a Study.
2. **Step 1: What do you want to calculate?: How many subjects do I need? (find sample size).** **Step 2: What kind of study?: Standard bioequivalence (80–125% limits, any region).** **Step 3: Study design: 2×2 standard crossover (TR | RT).**
3. Under Parameters, enter **Within-subject variability (CV %)** 25, **Expected Test/Reference ratio (%)** 95 and **Target power** 80. Click **Calculate**.

Questions: 1) How many subjects are needed? 2) Switch Step 1 to **What is the chance of success for my planned study? (calculate power)**, enter 30 as **Total number of subjects in your study**, and click **Calculate**. What is the power? 3) Switch back to the sample size, set the CV to 35 and click Calculate. How many subjects are needed now? A result is cleared when you change an input: click Calculate again after every change.

## Tutorial 6: Partial AUCs

Use the crossover data of Tutorial 4 (sampled at 0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8, 12 and 24 h).

1. In 5. All Subjects (Batch), enter Dose 100 (mg, h, ng/mL). Under **Partial AUCs (optional)**, set **Number of intervals** to 2. Interval 1: Start 0, End 1, and tick **Also report Cmax and Tmax in this interval**. Interval 2: Start 4, and tick **End at the last measurable concentration (t)**. Click **Run Analysis on All Subjects**.
2. Go to 6. Bioequivalence, with the settings of Tutorial 4. Each path has its own intervals, so enter them again in its Partial AUCs panel, with the Cmax box ticked for interval 1. Set **Role in bioequivalence** to **Pivotal (verdict)** for interval 1 and **Supportive (no verdict)** for interval 2. The partial metrics appear under **Parameters to compare**, already ticked. Click **Run Complete BE Analysis**.

Questions: 1) Work out Partial AUC 0–1 of subject 1 under Test by hand, from the first four samples (0, 12.3, 45.6 and 89.2 ng/mL), and compare it with the batch table. 2) What are the ratio and 90% CI for Partial AUC 0–1? 3) What does the table show for Partial AUC 4–t, and why? 4) What is the ratio for Cmax 0–1?

## Tutorial 7: Figures

These three examples use Visualize Data (card 3). No analysis settings are needed.

**A quick look after upload.** Upload example\_theoph.csv (Tutorial 3) and open 3. Visualize Data. The individual profiles plot shows 12 lines, one per profile, on a linear axis. Switch **Y-axis scale** to Log<sub>10</sub>: a straight tail points to mono-exponential elimination.

**A summary plot for a crossover.** Upload example\_be\_crossover.csv with the Treatment column mapped (Tutorial 4), open Visualize Data and choose **Summary Plot**. It overlays one geometric mean curve per treatment. This is a visual check, not a test: only the confidence interval in Bioequivalence says whether the products are bioequivalent. For a report, go to **Export Figure** in the sidebar, choose PDF, 7 × 5 in and 300 DPI, and click **Download Figure**.

**Dose-normalised profiles.** With example\_theoph.csv, whose Dose column differs per subject, tick **Normalise to dose (show C/Dose)** and colour the lines by Subject. Change the Y-axis label to "C/Dose (mg·L<sup>-1</sup> per mg/kg)". With dose-proportional pharmacokinetics the normalised profiles overlap.

## Answer Key

**Tutorial 1.** 1) C<sub>max</sub> 28.3 ng/mL at 1 h. 2) 156.8 ng/mL·h. 3) 4.607 h, from the last 5 points (adjusted R<sup>2</sup> 0.9724). 4) 616.7 L/h: 100 mg divided by an AUC to infinity of 162.2 ng/mL·h, converted to litres.

**Tutorial 2.** 1) Subjects 2 and 5 (0.5 h), and 4 and 7 (0.25 h). 2) 24 h: under Rule 1 the BLQ result at 8 h becomes 0, but the measurable values after it still count, and the fall to zero adds its area to AUC. 3) 2.238 ng/mL·h under Rule 1 and 2.394 under Rule 6. Rule 6 sets the BLQ samples during the lag to LLOQ/2, which adds area before the first measured concentration. The lag time stays 0.5 h, because a value set by a BLQ rule is not a measured concentration.

**Tutorial 3.** 1) 8.759 mg/L. 2) 98.65 mg/L·h. 3) 1.135 h. 4) Subject 1, 14.3 h. 5) 28.43%. 6) Not very: it rests on 3 points, and 31.5% of AUC to infinity is extrapolated (flagged in amber). Theophylline's half-life is usually about 8 h; report this value with a flag, or leave it out of the descriptive statistics as your analysis plan says. 7) Subjects 1 (0.74 mg/L, 7% of its C<sub>max</sub>), 7 and 10. Drug before the dose means carry-over from earlier dosing or an endogenous level; in a bioequivalence study ICH M13A excludes a period with a pre-dose value above 5% of C<sub>max</sub> from the primary analysis. The single dose of 320 mg is also not right for every subject: subject 9 received 3.10 mg/kg × 86.4 kg = 268 mg, so its CL/F is about 19% too high.

**Tutorial 4.** 1) 108.62% (106.27–111.02%). 2) 111.25% (108.95–113.60%). 3) Yes, both intervals lie within 80.00–125.00%. The note says that the study has 6 evaluable subjects, fewer than the 12 ICH M13A asks for, so the verdict is a statistical result only: this study would not be accepted as a pivotal study. 4) The residual variance is 0.000315, so the CV is about 1.8%. The example data are simulated with very little noise.

**Tutorial 5.** 1) 28 subjects. 2) 83.4%. 3) 52 subjects.

**Tutorial 6.** 1) 42.48 ng/mL·h (1.5375 + 7.2375 + 33.7); the concentrations rise throughout, so both trapezoidal methods give the same value. 2) 113.55% (110.90–116.25%), bioequivalent. 3) 112.54% (110.10–115.05%) without a verdict, because the interval is supportive. 4) 110.69% (108.31–113.11%).

**Tutorial 8.** 1) Two runs on the same data, the second with "r2adj\_threshold: 0.7 → 0.9"; the label says there were 2 runs before the record. 2) The times and concentrations were typed in by hand, so the reviewer must check every value against the source document. 3) Three: all runs on the same data before the record are listed, including those that led to the rejected record. 4) The analyst's name and user ID with the UTC time of creation, and the meaning "I have checked the settings of this analysis against the protocol, its results and its audit trail, and I approve this record." 5) The Approve and Reject buttons do not appear for one's own record, and the server refuses such a signature.

## Preparing Your Data and Study Types

### General Format Rules

The app expects data in **long format**: one row per sample, with a subject identifier, the sampling time and the measured concentration. Accepted files are CSV (.csv), text (.txt, .tsv) and Excel (.xlsx, .xls). The app suggests a column mapping from the column names (Appendix C); check it before you process the data. When another column could also be the time or the concentration, a note names it. CDISC ADNCA datasets are uploaded differently, see Scenario 9.

**Time** is the time since the dose of that profile, with the pre-dose sample at 0. A negative pre-dose time is used as it is: the lag time then becomes negative and AUC includes the time before the dose, so set the pre-dose sample to 0.

**Actual or nominal times.** Use the actual sampling times for the NCA, as ICH M13A (2.2.2.2) asks: a sample taken at 1.1 h enters the calculation at 1.1 h. Nominal times (1 h) belong in the mean plot of Visualize Data, which averages per time point. Keep a second copy of the file with nominal times for that figure.

### Decimal Separators and Regional Formats

Choose the delimiter (comma, semicolon or tab) and the decimal point (period or comma) in the upload settings. With a decimal comma, values such as 12,5 are read correctly, also in a concentration column that contains BLQ text such as “<0,5”. Excel files need no settings, because Excel stores numbers as numbers. In a decimal-comma file a point can only separate thousands, so 12.500 is read as 12500. Any other point in a number is refused, and the message names the column.

### Units and Doses

Use one concentration unit, one time unit and one dose unit for the whole file, and select them in the app. They set the conversion factor for clearance and volume, so a wrong unit gives wrong CL/F and Vz/F while C<sub>max</sub> and AUC look normal. A unit column with more than one unit (even ng/mL and µg/L) is refused; convert the values first. Molar concentrations (for example nmol/L) with a mass dose (mg) need the molecular weight, which you enter in the analysis settings. When the file states its units in a unit column (such as ConcUnit or Time\_Unit, or the CDISC variables AVALU and RRLTU), the app pre-selects them and refuses a selection that contradicts them.

A Dose column must hold the amount given, in the dose unit you select. A weight-based dose in mg/kg with the unit set to mg gives clearance and volume per kilogram, labelled L/h and L. Convert mg/kg to mg first (dose × body weight), or report the values as per kilogram.

### Scenario 1: Single Subject

To upload a file, include a Subject column; it may hold the same value on every row. Without it the upload cannot tell which rows belong together, and the app picks another column as Subject. Manual entry in One Subject at a Time needs only Time and Concentration (Tutorial 1).

| Subject | Time | Concentration |
|---------|------|---------------|
|---------|------|---------------|

|    |     |      |
|----|-----|------|
| P1 | 0   | 0    |
| P1 | 0.5 | 4.2  |
| P1 | 1   | 18.5 |
| P1 | 2   | 12.1 |
| P1 | 4   | 6.3  |
| P1 | 8   | 2.8  |
| P1 | 24  | 0.4  |

## Scenario 2: Multiple Subjects, Same Dose

| Subject | Time | Concentration |
|---------|------|---------------|
| S001    | 0    | 0             |
| S001    | 1    | 22.5          |
| S001    | 24   | 0.4           |
| S002    | 0    | 0             |
| S002    | 1    | 28.1          |
| S002    | 24   | 0.6           |

## Scenario 3: Multiple Subjects, Different Doses

Add a Dose column if subjects received different doses (dose escalation, weight-based dosing), or if a subject received different doses in different periods. The app takes the dose of each profile (subject, treatment and period) from this column, so every period is analysed with its own dose. For dose proportionality, see Scenario 11.

## Scenario 4: Standard 2-Period Crossover (BE)

Six columns: Subject, Treatment, Period, Sequence, Time, Concentration. Time restarts at 0 in each period: it is the time since the dose of that period, not since the first dose of the study (the app refuses profiles that do not start near time zero). The ratio is Test ÷ Reference. In Bioequivalence you choose which treatment is the Reference; the app pre-selects it only when the name is unambiguous (R, Ref, Reference, Comparator, Innovator, Originator or RLD). Map the Period column. Without it a crossover is still analysed but gets no verdict, because the period effect cannot be separated from the treatment effect. Columns named APERIOD, Occasion or OCC are recognised as Period. A Visit column is not, so map it yourself if it holds the period. Subject IDs must be unique in the whole study: if each sequence numbers its subjects from 1, different people would be merged into one subject, so the app refuses the file.

If the protocol asks for a partial AUC, take a sample at every cutoff time. A cutoff between two samples is interpolated, so its value then depends on the trapezoidal method. The interval itself comes from the protocol or the product-specific guidance, never from the data.

| Subject | Treatment | Period | Seq | Time | Conc |
|---------|-----------|--------|-----|------|------|
| S01     | Test      | 1      | TR  | 0    | 0    |

|     |           |   |    |   |      |
|-----|-----------|---|----|---|------|
| S01 | Test      | 1 | TR | 1 | 24.5 |
| S01 | Reference | 2 | TR | 0 | 0    |
| S01 | Reference | 2 | TR | 1 | 22.8 |

### Scenario 5: Replicate Designs and Paired Comparison

Same six-column format. Replicate designs: 2×2×3 full replicate (sequences TRT and RTR), 2×3×3 partial replicate (TRR, RTR, RRT) and 2×2×4 full replicate (TRTR, RTTR). Period goes to 3 or 4 and a treatment appears more than once per subject; each administration is analysed as its own profile. Paired comparison: all subjects received the treatments in the same order, so Sequence has one value (or can be left out).

### Scenario 6: Parallel-Group BE

Four columns: Subject, Treatment, Time, Concentration. No Period or Sequence needed.

### Scenario 7: Steady-State PK

Same layout as Scenario 2, with one dosing interval per subject. Time 0 is just before the dose of that interval, and the pre-dose concentration is above zero. Include the pre-dose sample and sample up to the end of the dosing interval ( $\tau$ ). Tick “Steady-state (drug given repeatedly)” and enter  $\tau$  in the app. If a profile has no pre-dose sample at time 0, the app assumes 0 there and says so, because  $AUC_{\tau}$  is then too low.

Two concentrations describe the low point of the interval.  $C_{min}$  is the lowest observed concentration from 0 to  $\tau$ , the pre-dose sample included; with an absorption lag it can fall after the dose.  $C_{\tau}$  is the observed concentration at  $\tau$ , and is left empty when no sample was taken at  $\tau$ . ICH M13A lists both for steady-state studies, and the EMA modified-release guideline asks for  $C_{\tau}$ . Both can be compared in Bioequivalence.  $AUC$  to infinity, its % extrapolated and the volume at steady state are single-dose quantities and are left empty.

A partial  $AUC$  at steady state has to lie within 0 and  $\tau$ .  $AUC_{\tau}$  is completed to  $\tau$  with  $\lambda_z$  when the last sample comes earlier; a partial  $AUC$  you enter yourself is not, so it stays empty for such a profile.

### Scenario 8: Drug–Drug Interaction Study

A drug interaction (DDI) study measures the effect of a co-administered drug (the perpetrator) on the pharmacokinetics of the victim drug. Most are crossovers in which each subject receives the victim drug alone and with the perpetrator, or fixed-order studies in which the victim is given alone first and with the perpetrator after a washout.

Use the same six columns as a crossover BE study (Scenario 4). Label the victim drug alone as “Reference” and the victim drug with the perpetrator as “Test”, or choose the Reference treatment in the app. The ratio is then (with perpetrator) ÷ (alone), the fold-change in exposure. For a single-dose victim, add  $AUC$  to infinity to the parameters to compare, as ICH M12 (5.1.1) asks; at steady state compare  $AUC_{\tau}$ ,  $C_{max}$  and  $C_{min}$  (ICH M12, 5.1.1). Analysing Drug–Drug Interaction Studies, in the chapter on using the app, explains how to read the result.

## Scenario 9: CDISC ADNCA Dataset

If your data come as a CDISC ADNCA dataset (columns such as USUBJID, AVAL, PARAMCD, NRRLT, ARRLT and ANL01FL), do not convert them by hand. The steps:

1. On the Upload page, set What kind of file? to CDISC ADNCA dataset and upload the .csv or .xlsx file.
2. Read the summary, and choose the time to use: ARRLT or MRRLT (actual times) for the NCA, NRRLT (nominal times) for a mean figure; AFRLT, time since the first dose, is refused. With several analytes or matrices, choose the one to analyse; Set pre-dose times (negative ARRLT) to 0 does what it says.
3. Click Convert dataset. The app applies ANL01FL, drops records with PCSTAT "NOT DONE", copies PCLLOQ into the LLOQ field, and refuses datasets it cannot convert safely, such as derived records (DTYPE), several units, or AVAL missing without a BLQ result.
4. Check the column mapping and the LLOQ, then click Process Data.

The choices are stored in the Analysis Record. SAS transport (.xpt) files are not read: converters/ADNCA\_TO\_FLAT.md in the repository gives a two-line R recipe to convert them to CSV. The same conversion is available outside the app as converters/adnca\_to\_flat.R.

## Scenario 10: IV Bolus and IV Infusion

For an IV bolus, choose IV Bolus (injected into vein at once) as the route. The concentration at time 0 is C0, the value just after the injection, not a pre-dose value: the app sets aside samples at or before time 0 and back-extrapolates C0 from the first two concentrations when they decline. Clearance and volume are reported without /F. AUC % Back-Extrapolated shows how much of AUC lies before the first sample; above about 20% the AUC is unreliable.

For an IV infusion, time 0 is the start of the infusion, and you enter the Infusion duration in the time unit of the data. One duration applies to every profile in a run, so split a file with infusions of different lengths and analyse each part separately. MRT is corrected for the infusion time (minus half the duration). Bioequivalence does not take an infusion.

## Scenario 11: Single Ascending Dose and Dose Proportionality

Give every subject a Dose column and put the dose level in a Treatment column (for example "10 mg", "30 mg", "100 mg"). All Subjects (Batch) then gives summary statistics per dose group. Tick Calculate dose-normalised parameters to get Cmax, AUC and the partial AUCs divided by each profile's dose, and use Visualize Data with Normalise to dose (show C/Dose) to compare the profiles. Overlapping dose-normalised profiles suggest dose proportionality but do not test it. The power model ( $\ln AUC = a + \beta \cdot \ln \text{Dose}$ , with a confidence interval for  $\beta$ ; Smith et al., 2000) is not in the app; fit it with other software on the exported table.

## Scenario 12: Multiple Dosing and Accumulation

A multiple-dose file usually holds a first-dose profile (Day 1) and a steady-state profile (Day N) for every subject. Code the day as Period, with time reset to 0 at each of the two doses,

and the day also in a Treatment column (for example "Day 1" and "Day 7"). The steady-state setting applies to every profile in a run, so:

1. For the Day N parameters, analyse the Day N profiles as steady state, with  $\tau$ .
2. For the accumulation ratio, use Bioequivalence with Paired comparison, Day N as Test and Day 1 as Reference, Steady-state ticked with the same  $\tau$ , and compare AUC Within Dosing Interval. Day 1 then gives AUC from 0 to  $\tau$  after the first dose, and the ratio with its confidence interval is the accumulation ratio. The note about a missing pre-dose sample does not apply to Day 1.
3. Document that steady state was reached with consecutive trough samples, as ICH M13A (2.2.2.2) asks.

### Scenario 13: Food Effect

A two-way food-effect study is a 2×2 crossover: the fed administration as Test and the fasted one as Reference (Scenario 4). Compare C<sub>max</sub>, AUC to last point and AUC to infinity, and report the median difference in T<sub>max</sub>. A 90% confidence interval within 80.00–125.00% supports the absence of a food effect. A three-way study (for example Test fasted, Test fed and Reference fasted) has three treatments, see Studies with Three or More Treatments.

### Scenario 14: Metabolites and Several Analytes

Analyse one analyte per run. Filter a flat file to one analyte first, or choose it during the ADNCA upload. A metabolite-to-parent ratio needs both in molar units: enter the molecular weight of each in its own run, and compute the ratio from the exported tables.

### Scenario 15: Absolute and Relative Bioavailability

Bioequivalence compares two treatments with one route and without dose normalisation, so it cannot compare IV with oral dosing, or strengths with different doses. Run All Subjects (Batch) per route with dose normalisation, and compute  $F = (\text{AUC oral} / \text{dose oral}) \div (\text{AUC IV} / \text{dose IV})$  from the exported table. For a confidence interval of F, use other software.

### Studies with Three or More Treatments

Bioequivalence takes exactly two treatments. For a three-way crossover (food effect, a three-arm DDI study, a Williams design), filter the file to two treatments per comparison and analyse each pair on its own. The residual variance then comes only from the periods of those two treatments, not from the ANOVA of the whole design that a statistical analysis plan usually specifies, so the intervals differ from a full-design analysis. When the plan asks for the full model, use other software.

### BLQ Handling

| Rule             | In the app                                   | What it does                                                                     | When to use it                                              |
|------------------|----------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------|
| Rule 1 (default) | Standard: pre-first = 0, post-last = missing | BLQ before the first and between measurable values = 0; after the last = missing | The usual convention for NCA and bioequivalence (see below) |

|        |                                                                  |                                                                                                                                     |                                                                        |
|--------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Rule 2 | All BLQ → zero                                                   | Every BLQ = 0                                                                                                                       | When the protocol sets all BLQ values to zero                          |
| Rule 3 | All BLQ → excluded                                               | Every BLQ = missing; the curve then starts from (0, 0) and runs across the removed samples                                          | Sensitivity analyses                                                   |
| Rule 4 | Before and between measurable → LLOQ/2, after the last → missing | BLQ before and between measurable values = LLOQ/2; after the last = missing; pre-dose = 0                                           | Sensitivity analyses; needs a justification in the protocol            |
| Rule 5 | Before peak → zero, after peak → excluded                        | BLQ before C <sub>max</sub> = 0; after C <sub>max</sub> = missing                                                                   | When the protocol specifies it                                         |
| Rule 6 | Before first quantifiable → LLOQ/2, rest → zero                  | BLQ after dosing and before the first measurable value = LLOQ/2; pre-dose, later BLQ and a profile without any measurable value = 0 | When the protocol imputes during absorption; it adds area during a lag |

Every assay has a lower limit of quantification (LLOQ): the lowest concentration it can measure reliably. Samples below it are reported as BLQ (below the limit of quantification). The app recognises “<0.5” (also “<0,5”), BLQ, BQL, BLOQ, ND (not detected) and NQ (not quantifiable), in any capitals. Set the LLOQ above 0 so the chosen rule is applied to these entries; each rule works per profile, in time order. NS (no sample), N/A and “missing” are treated as missing values, not as BLQ. If you click Process Data with the LLOQ left at 0, the app stops; when the entries look like “<0.5”, it shows the value it found and a button Set LLOQ to 0.5, which fills in the field (Tutorial 2).

ICH M13A (2.2.2.2) treats BLQ values as zero in PK parameter calculations and leaves them out of the half-life. The app never uses a value set by a rule for the half-life or the lag time. Rule 2 sets every BLQ value to zero. Rule 1 does so before and between measurable values but leaves values after the last one missing, which gives the same AUC to last point. With Rules 3 to 6 the Bioequivalence page reminds you that the protocol has to justify the rule.

In the processed data the app adds a BLQ\_flag column marking every value the rule set. It appears in the data preview, and the partial AUC notes and the bioequivalence table use it to report how much of a result rests on such values. A column of that name in your file is replaced.

The rule matters most for partial AUCs. An early window often holds BLQ samples, which Rule 1 sets to 0 and Rule 6 to LLOQ/2, and a late window over a plateau near the LLOQ can rest mainly on values the rule produced. The app says so when more than half of the samples in a window come from the BLQ rule, or when fewer than three measurable concentrations are left in it.

## Common Mistakes

**Wide format:** subjects as columns instead of rows. **Time not reset:** in a crossover, time restarts at 0 in every period. **A negative pre-dose time:** set it to 0. **0 for a missing sample:**

leave the cell empty, or write NS. **Duplicate time points:** usually several analytes, matrices or periods in one column; filter the file, or map the Treatment and Period columns. **LLOQ left at 0:** processing is blocked when BLQ text is present. **Dates or clock times as time:** the Time column must be elapsed time since the dose. **Mixed or wrong units:** select the units that match the file; they drive the clearance and volume conversion. **Dose in mg/kg with the unit set to mg:** see Units and Doses. **Subject IDs that restart in each sequence:** give every subject its own ID.

**A cutoff with no sample near it:** the protocol asks for AUC 0–30 min, but the samples are at 15 and 60 min. The app interpolates the concentration at 30 min, so the value depends on the trapezoidal method.

**Excluding a subject by deleting rows:** the app has no control to exclude a subject or period, and the Analysis Record then holds only the edited file. Follow the procedure in Exclusions, Outliers and Deviations, in the chapter on regulated use.

## Using the App

This chapter describes each path, with the labels the app shows. The Tutorials chapter takes you through them step by step; Reading Your Results explains the tables.

### Path 1: Plan a Study

Calculates the sample size, or the power for a given number of subjects, with the PowerTOST package. The page has four cards. **Step 1: What do you want to calculate?** chooses between **How many subjects do I need? (find sample size)** and **What is the chance of success for my planned study? (calculate power)**. **Step 2: What kind of study?** chooses the method: standard average bioequivalence, highly variable drugs (EMA ABEL or FDA RSABE), or narrow therapeutic index drugs (FDA NTID). **Step 3: Study design** offers the designs of Bioequivalence except the paired comparison: 2×2, 2×2×3, 2×3×3, 2×2×4 and parallel groups for standard bioequivalence; the replicate designs for ABEL and RSABE; only 2×2×4 for NTID. **Parameters** holds the numbers. Click **Calculate**; a result is cleared when you change an input, so click Calculate again after every change.

The CV the planner asks for depends on the design and the method:

| Design and method                               | CV asked for                               | Field label                                                                   |
|-------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------|
| Crossover or replicate, standard bioequivalence | Within-subject CV                          | Within-subject variability (CV %)                                             |
| Parallel groups                                 | Total CV (between plus within subjects)    | Total CV, between + within subjects (CV %)                                    |
| ABEL, RSABE, NTID                               | Within-subject CV of Test and of Reference | Within-subject CV of the Test product (CV %) and of the Reference product (%) |

Enter the expected Test/Reference ratio (the default follows the method, as in PowerTOST: 95% for standard bioequivalence, 90% for highly variable drugs, 97.5% for narrow therapeutic index drugs) and, for sample size, the target power. The significance level (0.05) and, for standard bioequivalence, the acceptance limits can be changed under Advanced: acceptance limits & significance level. After a log-transformed bioequivalence analysis of your own data, a button such as Use the Peak Concentration (C<sub>max</sub>) within-subject CV from my BE analysis (1.8%) offers its CV; with several metrics, Metric from the BE analysis chooses which. It offers the within-subject CV of a crossover for a crossover design, the total CV of a parallel analysis for a parallel design, and CV<sub>wT</sub> and CV<sub>wR</sub> of a replicate analysis for the scaled methods. The spread of C<sub>max</sub> between subjects in batch results is not a within-subject CV, so the app does not offer it. The results are in three tabs: Power vs. Number of Subjects, Sample Size vs. Variability, and Calculation Summary.

For a pivotal partial AUC, plan with the within-subject CV of that partial AUC. An early window is often more variable than total AUC, so a CV taken from AUC gives too few subjects.

The planner makes no record. For a protocol, write down the method, design, CVs, expected ratio, power, significance level, the resulting N, and the versions of the app and PowerTOST (About page), and keep a screenshot of the Calculation Summary.

## Path 2: Upload & Check Data

Choose **What kind of file? (Simple table (one row per sample) or CDISC ADNCA dataset)**, upload it, and set the Delimiter and Decimal point for a text file. Check the column mapping (Subject ID, Time, Concentration, and where present Treatment / Formulation, Period, Sequence and Dose), set the **LLOQ value** and the **BLQ handling rule**, and click **Process Data**. The app runs more than 30 quality checks: errors block processing, warnings do not. Some checks stop data the app cannot analyse safely: a CDISC-style file uploaded as a simple table, mixed units, dates or clock times in the Time column, profiles that do not start near time zero (time since the first dose), several profiles stacked in one column (duplicate times), subject IDs that restart in each sequence, and a decimal point in a decimal-comma file. A second dose inside one profile, or a multi-period file without a mapped Period, gives a warning. Processing a new file clears all previous results. Afterwards a panel links to the next steps: Visualize Data, Single Subject, Batch Analysis and Bioequivalence. Reading Your Results explains the quality report.

## Path 3: Visualize Data

Creates concentration–time figures from the processed data, from the hub or with the Visualize Data button on the Upload page. The settings are in the sidebar on the left; the plots are stacked on the right, and the **Plot Type** card decides which plots appear.

### Plot Types

**Plot Type** has three options. Individual Profiles (spaghetti plot) draws one concentration–time line per profile (subject, treatment and period), coloured by a grouping of your choice; use it to look at individual variability, outliers and carry-over. **Summary Plot** (geometric mean  $\times/\div$  geometric SD) shows the geometric mean at each time point, with error bars from the geometric mean divided and multiplied by the geometric SD. Treatment groups are overlaid automatically. Both plots shows the spaghetti plot above the summary plot.

Concentrations at a time point are usually skewed to the right, so the geometric mean describes the typical profile better than the arithmetic mean. The **Summary statistic** dropdown also offers the arithmetic mean  $\pm$  SD, labelled as arithmetic and with a warning. Values of zero, for example BLQ results set to 0, cannot enter a geometric mean and are left out at that time point, so at time points where most samples are BLQ the mean is biased upwards; the figure legend counts the values left out.

### Sidebar Controls

**Y-axis scale** (Linear or Log<sub>10</sub> (semi-log)): the semi-log view makes the terminal elimination phase linear, which shows biphasic profiles and half-life problems. Zero concentrations cannot be shown on a log scale; they are omitted with a note.

**Colour lines by** (spaghetti plot): Subject ID (always available), Treatment, Period and Sequence (when that column was mapped), and None. In a crossover, colour by Period to look for period effects, or by Treatment to compare formulations. **Show data points on lines** adds the samples to the spaghetti plot; **Overlay individual observations** adds them to the summary plot. **X-axis tick spacing** sets the distance between time labels (Auto by default).

**Normalise to dose (show C/Dose)**, a checkbox that appears only when a Dose column was mapped: concentrations are divided by the dose, so the Y axis shows C/Dose (for example ng/mL per mg). A missing or zero dose gives a missing value. Change the Y-axis label to match, for example “C/Dose ( $\text{ng}\cdot\text{mL}^{-1}\cdot\text{mg}^{-1}$ )”.

**Shade the partial AUC intervals of the last analysis** shades on the summary plot the intervals of the last batch or bioequivalence run; the Figure Record reproduces the shading.

**Colour palette:** Default (app palette), Colorblind-safe (Okabe-Ito), or Grayscale. For print, choose Grayscale and export at 600 DPI. **Theme:** Clean (white background), recommended for reports, Minimal, or Classic. The X-axis label, Y-axis label, Plot title and Base font size (pt) (8–18) can be edited. The default Y-axis label is “Concentration”; add the drug and unit, for example “Cyclosporin (ng/mL)”. The default X-axis label is “Time (h)”.

### Suggested Figure Legend

Above the plots, a card titled Suggested figure legend holds a ready-to-copy description of the current figure: number of subjects and observations, plot type, summary statistic, error bars, treatment names, the number of concentrations  $\leq 0$  left out of the geometric mean, **Y-axis scale** and colour grouping. It updates when you change settings. Paste it into your report and add the drug name and units.

### Exporting a Figure

Export controls are at the bottom of the sidebar under **Export Figure**, always visible. A small “Will export” label above the button shows which plot will be downloaded; with Both plots, Which plot to export? chooses one. Format: PNG (Word documents, presentations, email), PDF (vector; recommended for journals that accept PDF figures), or SVG (to edit in Inkscape or Adobe Illustrator). Width (in) and Height (in) take 2–20 inches; journals usually ask for 3.0–3.5 in for a single column and 6.5–7.0 in for a double column. Resolution (DPI): 150 (screen and drafts), 300 (print, default) or 600 (high-resolution submission). **Download Figure** is grey until the first plot has appeared. The file name includes the plot type (spaghetti\_plot.png or summary\_plot.png), so both can be downloaded without overwriting each other.

The plots on screen are interactive (plotly): hover to see subject, time and concentration; drag to zoom; reset with the home icon. The exported file is a separate static figure (ggplot2) built from your sidebar settings.

## Path 4: One Subject at a Time

Under **Where is your data?**, choose **Use my uploaded dataset (from step 2)** to step through the profiles with the arrow buttons, or **Enter data manually (type values directly)** to type times and concentrations, one per line, and click **Use This Data** (at least 3 points; Tutorial 1). Under **Analysis Settings**, set the Dose (filled in from the Dose column when there is one), the units, the **Route of administration** and the minimum  $R^2$ , and click **Run PK Analysis**. The **PK Parameters** card shows the results; the **Half-Life Calculation** card shows the terminal-phase points, which you can change under **Points for half-life:** and **Recalculate**. The result belongs to the profile and settings it was computed with: selecting another profile, or changing a setting, clears it. Downloads: **Download Results (CSV)** and a Complete Analysis Record.

On a controlled installation, data you type in are marked in the audit trail and on the Records page, and the reviewer checks every value against the source document.

**Partial AUCs (optional):** set **Number of intervals** and enter the start and end of each, in the time unit of your data. Tick **End at the last measurable concentration (t)** for an interval that runs to the end of the profile, and **Also report Cmax and Tmax in this interval** when the protocol asks for the peak inside a window. An interval that reaches past the last measurable concentration stays empty, with a note saying so: partial AUCs are never extrapolated.

## Path 5: All Subjects (Batch)

The sidebar has three cards. **Administration:** **Route of administration**, **Dose information (Same dose for all subjects, or Doses differ by subject or period (from Dose column in data))**, the dose, the infusion duration for an infusion, and **Steady-state (drug given repeatedly)** with the Dosing interval  $\tau$ . **Units:** Dose unit, Time unit, Conc unit, and the molecular weight for molar units. **Calculation Method:** **Trapezoidal method (Linear-up / Log-down**, the default, or **Linear-up / Linear-down)**, **Minimum R<sup>2</sup> for half-life estimation** (default 0.70), and **Calculate dose-normalised parameters**. Then **Partial AUCs (optional)** and **Run Analysis on All Subjects**.

The results are in five tabs. Concentration-Time Profiles shows all profiles and a Mean  $\pm$  SD plot; that mean is arithmetic, so use Visualize Data for geometric summaries. **All Parameters (per profile)** is the table, with a short default view and **Show all parameters** for every column; AUC % Extrapolated above 20% is flagged in amber. **Summary Statistics** gives the statistics per treatment. Individual Profiles (grid view) shows one small plot per profile, and **Half-Life Review** lets you choose the terminal-phase points of any profile: select it under Profile:, tick the points under **Points for half-life:** and click **Recalculate**. A selection stays with its profile; **Remove override (automatic fit)** returns it to the automatic fit. A CDISC parameter codes panel lists the official code of each parameter.

When the automatic terminal fit of a profile has an adjusted R<sup>2</sup> below the minimum, its half-life, AUC<sub>∞</sub>, CL/F, Vz/F and MRT are left empty and a note names the profile; select the points in **Half-Life Review** if you want a value. Profiles with fewer than 2 positive concentrations are left out, with an amber note listing them. A single-dose run also notes profiles whose pre-dose concentration is above 5% of C<sub>max</sub>. At steady state, parameters extrapolated to infinity and the single-dose volume and MRT are left empty. Changing a setting clears the results, so run the analysis again.

Partial AUCs join the table, the summary statistics and the downloads. An interval that reaches past the last measurable concentration of a profile stays empty for that profile, and a note names the profile and its last measurable time. With dose normalisation switched on, the partial AUCs and the C<sub>max</sub> within an interval are divided by the dose as well; the time of that maximum is not.

Downloads: **Download Results (CSV)**, **Download Results (Excel)**, with summary statistics per treatment and a sheet of CDISC codes, and a Complete Analysis Record. For a study report, use the record: the plain downloads carry no app version, data hash or settings.

## Path 6: Bioequivalence

Runs the NCA, then the ANOVA, then the confidence interval and a forest plot, and gives a verdict where one is justified.

### Step 1: PK Analysis Settings

**Dose information**, dose, units, **Route of administration** (oral, IM or SC, or IV bolus; an IV infusion is not supported here), trapezoidal method and minimum  $R^2$ , as in Path 5, and **Steady-state (drug given repeatedly)** with  $\tau$  for steady-state data. At steady state AUC Within Dosing Interval replaces AUC to last point and AUC to infinity; after a first run, Minimum Concentration (Cmin) and Concentration at Tau (Ctau) can be added to the parameters to compare.

### Step 2: Bioequivalence Settings

**Study design**: 2×2 standard crossover, 2×2×3 or 2×2×4 full replicate, 2×3×3 partial replicate, parallel groups, or paired comparison. The design found at upload is pre-selected, and a crossover chosen for one-period data is flagged. If all subjects received the treatments in the same order, the data are analysed as a paired comparison.

**Reference treatment**: the ratio is Test ÷ Reference. The run is blocked until one is chosen, and the results table shows the comparison, for example “New / Old”.

**Statistical model**: leave it at **Fixed effects (all terms fixed)** unless the protocol specifies the mixed model. The model has Sequence, Subject, Period and Treatment as factors; with fixed effects this is EMA Method A, and the mixed model (random subject intercept) is EMA Method B, not the FDA model for replicate designs. Fixed-Effects vs. Mixed-Effects Models, in Statistical Methods, explains the difference.

**Log-transform (recommended)**, **Parameters to compare** (Cmax and AUC to last point by default, the primary parameters of ICH M13A; AUC to infinity, Tmax and half-life can be added), **Confidence interval** (80 to 99%; a verdict is given only at 90%) and the acceptance limits **Lower (%)** and **Upper (%)**. Click **Run Complete BE Analysis**. Changing a setting clears the results; the parameters you chose are kept.

### Widened and Narrowed Limits

With limits wider than 80–125%, two more settings appear. **Widened limits apply to** chooses Cmax only, Cmax and partial AUCs (not AUC to last point or to infinity), or All compared metrics (e.g. no-effect boundaries); the default is Cmax only. Also require the point estimate within 80.00–125.00% is ticked; untick it only for comparisons without such a rule, such as drug-interaction no-effect boundaries.

**Widened limits are not ABEL or RSABE**. Entering limits wider than 80.00–125.00% does not perform a reference-scaled analysis: the app does not check the conditions for widening (a pre-specified, clinically justified widening, CVwR of this study above 30%, the cap at 50%, a CVwR not driven by outliers; EMA 2010, section 4.1.10), and it does not compute the FDA scaled criterion. Use widened limits only when the protocol fixes them in advance, and document the justification outside the app. For replicate designs a table shows the within-subject variability of the Reference (and of the Test when replicated) and the EMA widened limits it would imply, for information only.

For a narrow therapeutic index drug under the EMA approach, enter 90.00 and 111.11. Narrowed limits apply to every compared metric; the FDA NTID scaled test and its variability comparison are not available.

## Running and Reading the Results

The tabs are **Test vs. Reference Comparison** (the table and the forest plot), PK Parameter Table, Statistical Details (ANOVA), Concentration-Time Profiles, Individual Profiles and **Half-Life Review**. Downloads: **Download Complete BE Report (Excel)**, **Download CI Table (CSV)** and a Complete Analysis Record. The verdict uses confidence limits rounded to two decimals. A design summary and an amber warning for subjects missing a treatment appear after the NCA. Reading Your Results explains the columns.

With the results the app lists points that ICH M13A asks you to check. A profile with fewer than 2 measurable concentrations has no NCA result and is counted under Profiles missing; its subject leaves the comparison, which M13A (2.2.1.1) accepts only as a planned exception, in general for no more than one subject. A period whose AUC to last point is below 5% of the geometric mean of that treatment is flagged as very low exposure, on the same terms. A single-dose study also lists a pre-dose concentration above 5% of  $C_{max}$ , and AUC to last point covering less than 80% of  $AUC_{\infty}$  in more than 20% of the profiles; neither check runs at steady state. Fewer than 12 evaluable subjects is checked on the smaller of the Test and Reference counts.

**A verdict is computed on all the data you uploaded.** When the checks name a period with a pre-dose concentration above 5% of  $C_{max}$ , ICH M13A (2.2.3.3) excludes it from the primary analysis, so the verdict shown is not the M13A primary analysis: exclude the period as the protocol says (see Exclusions, Outliers and Deviations) and run again. With fewer than 12 evaluable subjects, M13A (2.2.3.1) does not accept the study as pivotal, and the verdict is a statistical result only.

For parallel groups the app also calculates the Welch interval, which does not assume equal variances, and reports it when it would change the conclusion. The pooled interval stays the primary result; if your statistical analysis plan or a guidance asks for an unequal-variance analysis, the app's verdict is not your primary result.

## When There Is No Verdict

The Bioequivalent? column says why.

| Situation                                                        | What the app shows                                                                                                             |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| A crossover without a mapped Period column                       | No estimate and no verdict; the reason says to map Period                                                                      |
| A confidence level other than 90%                                | The interval, without a verdict                                                                                                |
| The mixed model was chosen and cannot be fitted                  | The fixed-effects result for information, no verdict: changing the model after seeing the data is a post-hoc change (ICH M13A) |
| A paired comparison (a single treatment order)                   | Ratio and interval, no verdict: period and treatment cannot be separated                                                       |
| Half-life, $T_{max}$ , or a parameter without log transformation | A ratio (half-life) or a difference, no verdict: not bioequivalence endpoints                                                  |
| A supportive partial AUC                                         | Ratio and interval, no verdict                                                                                                 |
| A metric with a value of zero in any profile                     | No estimate and no verdict, with the zero counts per treatment                                                                 |

Tmax is compared as an untransformed difference with a t-based interval. Tmax is discrete and is usually reported as median and range, with a nonparametric test when the plan asks for one; do that outside the app, and ignore geometric statistics for Tmax.

### Partial AUCs in Bioequivalence

Partial AUC intervals here also carry a role, set under **Role in bioequivalence**. A **Pivotal (verdict)** interval receives a bioequivalence verdict, a **Supportive (no verdict)** interval only its ratio and confidence interval. The metrics appear under **Parameters to compare**, ticked, as soon as the intervals are valid, and Cmax within an interval can be compared as well. A metric whose value is zero in any profile receives no estimate and no verdict, because zero cannot be log-transformed and the profiles that would drop out are the low-exposure ones. That metric keeps its row in the table and in the forest plot, with the reason, the zero counts per treatment and the number of profiles that have no value.

## Analysing Drug–Drug Interaction Studies

Bioequivalence (Path 6) can also analyse drug–drug interaction (DDI) studies: the design compares two conditions within subjects, with an ANOVA model and a confidence interval for the geometric mean ratio.

### How to Set Up a DDI Analysis

Prepare the data as a crossover dataset (Scenario 4 or 5). Choose the victim drug given alone as the **Reference treatment**, so the ratio is (with perpetrator) ÷ (alone): the fold-change in exposure. For a single-dose victim, tick AUC to infinity under **Parameters to compare** as well; at steady state compare AUC Within Dosing Interval, Cmax and Cmin.

Run the analysis. A ratio of 200% means the perpetrator doubled the victim's exposure; 50% means it halved it. Check the Comparison column to confirm which treatment is in the denominator.

### Interpreting DDI Results

A DDI study quantifies how large the interaction is; it is not a test of equivalence. ICH M12 (section 5.2) asks for the geometric mean ratio with and without the precipitant and its 90% confidence interval for AUC and Cmax (and Cmin when relevant), and for the range of individual ratios. The app does not give individual ratios: compute them from the exported PK Parameter Table. Interpret the result against no-effect boundaries for the victim drug, preferably based on its exposure-response relationship (section 5.3.1). The default no-effect boundaries are a 90% CI within 80–125%, which M12 calls generally acceptable but overly conservative for most drugs.

If the study used a sensitive index substrate, the point estimate also classifies the precipitant (M12 section 5.3.2): a strong, moderate or weak inhibitor raises the substrate's AUC at least 5-fold, 2- to 5-fold, or 1.25- to 2-fold; a strong, moderate or weak inducer lowers it by at least 80%, by 50 to 80%, or by 20 to 50%. These classes describe the precipitant, not the size of the interaction for other victim drugs.

### Practical Considerations

To compare with no-effect boundaries, enter them as acceptance limits, choose All compared metrics under **Widened limits apply to**, and untick the point-estimate

requirement if it does not apply. A YES in the Bioequivalent? column then means that the 90% CI lies within the boundaries you entered; it is not a bioequivalence conclusion.

For fixed-order DDI studies (all subjects receive the victim alone first, then with the perpetrator), select Paired comparison. The app reports the ratio and its confidence interval without a verdict, because period and interaction effects cannot be separated: read the interval against the boundaries yourself.

The forest plot and confidence interval table can go into the DDI section of a clinical study report when they were produced on a qualified installation and reviewed according to your procedures.

## Complete Analysis Record

Available from One Subject at a Time, All Subjects and Bioequivalence, with **Generate Analysis Record**; Visualize Data produces a similar Figure Record. Fill in **Analyst name (optional)** and **Study name (optional)** first: they appear in the record, but they are typed, not signed in, so they do not establish who did the analysis. On a controlled installation, the analyst is your name from the sign-in, and the record is also stored on the server for review (Working on a Controlled Installation).

The zip contains results.xlsx, app\_results\_reference.csv, analysis\_settings.json, nca\_pipeline.R (the app's own data-processing code), reproduce\_analysis.R, reproduction\_check.txt, data\_integrity.txt, analysis\_summary.html and a copy of the data file. From the batch and bioequivalence paths, results.xlsx has a Checks sheet, and the summary a Checks and Notes section, with the data quality findings and the notes shown with the results, such as the ICH M13A checks. data\_integrity.txt holds SHA-256 hashes of the data, the settings, the results, the reference results, the script and the pipeline code. A SHA-256 hash is a 64-character fingerprint that changes if a single character in the file changes.

When it creates the record, the app runs the reproduction script in a separate R process and writes the outcome to reproduction\_check.txt:

| Outcome      | Meaning                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| MATCH        | The shipped pipeline, run again on the enclosed data and settings, gives the same NCA parameters (largest relative difference below $10^{-6}$ ) |
| CLOSE        | The same within a relative difference of $10^{-3}$ , for example after a package update that changes rounding                                   |
| DIFFERENT    | Different results, or a data file, pipeline code or reference results that were changed, or a parameter present on one side only                |
| NOT COMPARED | The reference results (app_results_reference.csv) are missing                                                                                   |

**What the record shows and what it does not.** MATCH shows that the analysis can be reproduced with the same code and packages. It is not independent verification. For a bioequivalence record the script recomputes the NCA parameters only: the ANOVA, the

confidence intervals and the verdict are recorded in results.xlsx and are not recomputed, and the script does not check results.xlsx. The manifest is not signed: whoever changes a file can also recompute its hash. To make the record tamper-evident, store the zip, or its SHA-256, in a controlled system. The record is not an audit trail in the sense of 21 CFR Part 11 or EU GMP Annex 11; it documents one analysis. Timestamps come from the clock of the computer that ran the app (UTC on shinyapps.io), and the summary names the time zone. On a controlled installation, the stored copy on the server is covered by the audit trail, and the reviewer's signature is bound to the record's SHA-256 (The Signature Sheet, in the chapter Working on a Controlled Installation).

If the uploaded file was no longer available when the record was made, the record holds the table as the app read it, written as CSV, with a different name and hash; the settings and the Checks say so. Keep the original source file with the record.

Partial AUC intervals, their roles and the BLQ share used for the notes are recorded, and the reproduction script recalculates the partial AUCs. Change an interval in the settings file afterwards and the check reports DIFFERENT.

**Re-running a record.** Unzip it into an empty folder. In R, set the working directory to that folder (`setwd("path/to/folder")`) and run `source("reproduce_analysis.R")`. The script checks the data hash, recomputes the results and prints the outcome.

## Reading Your Results

### The Data Quality Report

After Process Data, the Upload page lists what the checks found, one line per finding.

| Column   | Meaning                                                                               |
|----------|---------------------------------------------------------------------------------------|
| Severity | ERROR blocks processing; WARNING and INFO do not; OK means the check passed           |
| Category | What was checked: Structure, Time, Concentration, Subjects, Design, Format and others |
| Message  | What the check found, for example "22 BLQ/text entries detected"                      |
| Detail   | The rows, subjects or values concerned                                                |
| Action   | What to do about it, where there is something to do                                   |

Fix every ERROR in the file or the mapping, and process the data again. Read the warnings too. They do not stop the analysis, but they can change what the results mean.

### The Parameter Table

All Parameters (per profile) in Path 5, and PK Parameter Table in Path 6, have one row per profile: a subject under one treatment in one period. Subject, Treatment and Period identify the row; the other columns are parameters, labelled in words with their unit, for example "AUC to Last Point (ng/mL·h)". The default view shows the main parameters; Show all parameters shows every column. Appendix A lists each parameter with its definition, unit and when it is reported. An empty cell is not an error: it means the parameter could not be estimated for that profile, and a note above the table says why (for example a terminal fit below the minimum adjusted  $R^2$ ).

### Summary Statistics

Summary Statistics gives, per treatment and parameter:

| Column                           | Meaning                                                                         |
|----------------------------------|---------------------------------------------------------------------------------|
| N                                | Number of profiles with a value                                                 |
| Geometric Mean                   | $\exp(\text{mean of } \ln \text{ values})$ ; the usual summary for AUC and Cmax |
| Geometric CV (%)                 | $100 \cdot \sqrt{\exp(\text{SD}^2 \text{ of } \ln \text{ values}) - 1}$         |
| Median, Q1 (25%), Q3 (75%)       | The middle value and the quartiles; the usual summary for Tmax                  |
| Arithmetic Mean, Std Dev, CV (%) | Mean, standard deviation, and SD as a percentage of the mean                    |
| Min, Max                         | The range                                                                       |

Geometric statistics need positive values; they are not meaningful for Tmax or for a parameter that can be zero.

## The Bioequivalence Table and Forest Plot

The table on the Test vs. Reference Comparison tab has one row per compared metric.

| Column                                                         | Meaning                                                                                                  |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| PK Parameter, Comparison                                       | The metric, and which treatment is divided by which ("Test / Reference")                                 |
| Subjects (Test), Subjects (Reference)                          | Subjects that enter the comparison                                                                       |
| Estimate                                                       | The geometric mean ratio Test ÷ Reference in percent (a difference for Tmax or untransformed parameters) |
| 90% CI Lower, 90% CI Upper                                     | The confidence interval, at the level you chose                                                          |
| Accept. Lower, Accept. Upper                                   | The acceptance limits used for this metric                                                               |
| PE within 80–125%                                              | Whether the point estimate lies within 80.00–125.00%, for widened limits                                 |
| Bioequivalent?                                                 | YES or NO, or the reason why there is no verdict                                                         |
| Profiles missing, Zero values, Mostly BLQ (Test and Reference) | Profiles without a value, with a value of zero, or resting mainly on values set by the BLQ rule          |
| Model                                                          | The model fitted, and whether a mixed model failed                                                       |

The forest plot shows each estimate as a point with its confidence interval as a horizontal bar, and the acceptance limits as vertical lines: a bar entirely between the lines is a YES. The Residual Variance (MSE) and Degrees of Freedom are in the downloads (Download CI Table (CSV)); the within-subject CV of a crossover is  $100 \cdot \sqrt{(\exp(\text{MSE}) - 1)}$ .

Statistical Details (ANOVA) shows, per metric, the model terms (Sequence, Subject, Period, Treatment) with their degrees of freedom and F tests. A significant Sequence or Period term is a reason to look at the data, not a reason to change the analysis.

## Judging the Terminal Phase

Half-life, AUC to infinity, CL/F, Vz/F and MRT all rest on the terminal slope  $\lambda_z$ . Before you report them, check each profile against these points, and follow your statistical analysis plan where it is stricter.

- **At least 3 points** in the terminal log-linear phase (ICH M13A, 2.1.8), after Tmax and not counting Cmax itself. The app accepts 2 points in a manual selection, but a 2-point fit has no measure of fit and should not be reported.
- **Span:** as a rule of thumb the points should cover at least two half-lives; ICH M13A (2.1.8) asks for sampling over at least three.
- **Adjusted R<sup>2</sup>:** the app's default minimum of 0.70 is a floor, not a recommendation; many analysis plans use 0.80 or 0.90.

- **AUC % Extrapolated** above 20% (amber in the table): the extrapolated part carries much of AUC to infinity. Report AUC to infinity, CL/F, Vz/F and half-life with a flag, or leave them out of the descriptive statistics, as the plan says.

Half-Life Review shows the fit the results use, on a log scale, with the chosen points marked.

## Checking Plausibility

- **Pre-dose concentration above zero:** carry-over, earlier dosing or an endogenous compound. In a bioequivalence study a value above 5% of C<sub>max</sub> takes the period out of the primary analysis (the app says so).
- **C<sub>max</sub> at the first sample after the dose:** absorption was faster than the sampling, and C<sub>max</sub> may be underestimated. ICH M13A (2.1.8.1) asks for this to be discussed.
- **T<sub>max</sub> at the last sample:** absorption was still going on, or sampling stopped too early; there is then no terminal phase and no  $\lambda_z$ .
- **A half-life much longer than expected:** check the number of points and the % extrapolated (Tutorial 3).
- **Flip-flop kinetics:** with slow absorption, such as a modified-release product, the terminal slope reflects absorption rather than elimination. Point selection cannot correct this: report the value as an apparent half-life. Only IV data can tell the two apart.
- **A profile without measurable concentrations** (non-absorber): it has no NCA result, and in bioequivalence it is counted as missing and named in the checks.
- **Very different results under two BLQ rules:** the result depends on the imputation; report the rule and consider a sensitivity analysis.

## Working on a Controlled Installation

Your organisation may run NCA Assistant on its own server in controlled mode. Then you sign in with a personal account, the app records what you do in an audit trail, your Analysis Records are stored on the server, and a reviewer signs them. This chapter is for analysts, reviewers and inspectors. Setting up the server is in Appendix E.

### What Is Different on a Controlled Installation

The analyses are the same. What changes is around them:

|                           | Public app and standard installation | Controlled installation                                                                                                                             |
|---------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Signing in                | Not needed                           | Personal user ID and password; a new account starts with the one-time password admin                                                                |
| Top bar                   | The usual buttons                    | Also a shield icon (hover over it to see the organisation and server), Records, Audit trail for reviewers and inspectors, and a menu with your name |
| Analyst name in a record  | Typed by you, optional               | Your name from the sign-in, not editable                                                                                                            |
| After you create a record | You download it                      | You download it, and a copy is stored on the server for review                                                                                      |
| Data typed in by hand     | Allowed                              | Allowed, and marked for the reviewer                                                                                                                |
| Audit trail               | None                                 | Every sign-in, data load, analysis, download, record and signature, with time and user                                                              |

**What is recorded about you.** The audit trail holds your user ID, name and role, and when you signed in, which data you loaded, which analyses you ran with which settings, what you downloaded and what you signed. Reviewers, inspectors and the system owner can read it. It is kept as long as the study records; your organisation's privacy notice says why.

### Signing In for the First Time

1. The system owner gives you your user ID and tells you to sign in at once. The starting password is **admin**.
2. Open the address of your installation and sign in with your user ID and admin.
3. The app asks for a new password. Use at least 12 characters, with a digit, a lower-case and an upper-case letter. The app refuses admin and anything shorter.
4. Sign in again with the new password. The top bar now shows your name.

If admin does not work for your new account, someone may have used it before you. Tell the system owner at once, so that the account is checked and reset.

## Changing Your Password

Open the menu with your name in the top bar and choose **Change password**. Enter your current password, then the new one twice. You stay signed in. The password also expires after 90 days, and the app then asks for a new one when you sign in.

Change it at once, and tell the system owner, when you think someone else knows it. If you have forgotten it, ask the system owner for a reset: you then sign in with admin and choose a new password.

After five wrong passwords at sign-in, your account is locked. Only the system owner can unlock it.

## Staying Signed In

After 15 minutes without activity you are signed out. Moving the mouse, typing and scrolling all count as activity. Two minutes before the end, a message counts down; click **Stay signed in**, or move the mouse. When you are signed out, the uploaded file, the settings and the results are gone, as when you close the browser: upload the file again. Records you created are safe, because they are stored on the server when you create them.

To sign out yourself, open the menu with your name and choose **Sign out**.

## Creating a Record for Review

Create an Analysis Record as usual, with **Generate Analysis Record** (Complete Analysis Record, in the chapter on using the app). The record names you as the analyst, from your sign-in. After the download, a line under the button says **Stored for review**, with the first characters of the record's SHA-256, and **Open in Records**. A Figure Record is stored in the same way but is not reviewed: its line says **Stored**.

On the **Records** page you see your own records and whether they are awaiting review, approved or rejected. A rejected record stays as it is. If the analysis must change, run it again and create a new record.

**Data typed in by hand.** In One Subject at a Time you can still type in times and concentrations. The record then says so, and the reviewer checks every value against the source document. The same applies to a file that someone typed from a paper source: tell the reviewer where the file came from.

## Reviewing and Signing a Record

**For reviewers.** Open **Records**. The list starts at **Awaiting review, not mine**; choose **All records** or **My records** to see others. Your own records have no Approve or Reject buttons: someone else must review them.

Select a record. The details show the type, the study, who created it and when, the outcome of the reproduction check, and the SHA-256. Below them, **Audit history of the data** lists every run on the same data before the record was made, with the settings that changed between runs. A label says so when there was more than one run. **Open summary** shows the record's own summary; **Download** gives the zip.

Before you approve, check:

- the settings match the protocol and the statistical analysis plan (design, Reference treatment, model, limits, BLQ rule and LLOQ, partial AUC intervals);
- the data SHA-256 matches the source file, and, where the data were typed in (the record is marked) or the file was typed from paper, every value matches the source document;
- earlier runs on the same data are explained, especially when a setting changed between runs;
- the reproduction check says MATCH;
- the checks and notes of the analysis have been dealt with.

Then click **Approve** or **Reject**. The dialog shows which record you sign and what your signature means. For a rejection, type the reason; **Sign** stays grey until you do. Type your user ID and password and click **Sign**, or press Enter. After a wrong password the message says how many attempts are left; the third failure signs you out. A record is reviewed once. After that it stays approved, or rejected.

## The Signature Sheet

Every reviewed record has a signature sheet, a web page you can print or file. It shows the record and its SHA-256, who created it and when, who reviewed it, when, with which meaning and reason, and the audit entry of the signature. All times are in UTC. **Download (with signature sheet)** gives a zip with the record and the sheet together; file both in the study file.

Before it shows a signature, the app checks that the stored record still matches its SHA-256. **Signature valid** means it does. **Signature INVALID**, in red, means the stored record has changed since it was signed: tell the system owner at once. To check a copy you hold, for example from the study file, upload it under **Verify a record file** on the Records page. The app lists the signatures of that exact file, or says that no record with its SHA-256 exists.

## Reviewing the Audit Trail

**For reviewers and inspectors.** Open **Audit trail**. The line at the top says when the trail was last reviewed and how many entries have been added since.

- **Exceptions** lists what needs attention: failed and locked sign-ins and security alerts, refused signatures and password changes, rejected records, stored records that no longer match their SHA-256, data analysed more than once before a record, data analysed without a record, app starts without a preceding stop, changes in the configuration, entries out of time order, activity outside working hours, and one user signed in twice at the same time. Start here.
- **All entries** shows the whole trail in plain words, newest first. Filter by date, user or event, or search for a file name or SHA-256. Select an entry to see all of it, including the settings of a run.
- **Users** lists every account with its roles, status and role history.

**Verify chain** checks that no entry has been changed, inserted or removed. **Export CSV** saves the entries of the current filter, for example one study, with the result of the check at the top. After a review, reviewers click **Sign trail review** and sign with user ID and

password; this records the period you reviewed. How often to review, and what else triggers a review, is set by your organisation's procedure.

## For Inspectors

An inspector account can see every record, its history and signatures, the audit trail and the user overview, and can export and verify, but cannot sign or change anything. The system owner creates one per inspection and deactivates it afterwards.

## Tutorial 8: From Analysis to Approved Record

Do this tutorial on a training installation, never on the production server: practice runs would end up in the real audit trail. A training installation in controlled mode runs on your own computer (Appendix E, A Training Installation). You need two accounts, an analyst and a reviewer; use two browsers, or a normal and a private window, to be signed in as both.

1. As the system owner, create an analyst and a reviewer account with `gxp/manage_users.R` add.
2. Sign in as the analyst and choose a new password. Open 4. One Subject at a Time, choose **Enter data manually (type values directly)**, click **Use This Data** and run the analysis with the settings of Tutorial 1.
3. Set **Min R<sup>2</sup> for half-life** to 0.9 and click **Run PK Analysis** again.
4. Type Tutorial 8 as Study name and click **Generate Analysis Record**. Read the line under the button.
5. Sign in as the reviewer in the other window. Open **Records**, select the record and read its details and the audit history of the data.
6. Click **Reject**, type the reason "Min R<sup>2</sup> changed from the protocol without a reason", and sign.
7. As the analyst, set **Min R<sup>2</sup> for half-life** back to 0.7, run the analysis again and create a new record.
8. As the reviewer, approve the new record, and download its signature sheet.

**Questions:** 1) What does the audit history of the first record show, and which setting changed? 2) Why is the data marked for the reviewer? 3) How many runs does the history of the second record list, and why? 4) What does the signature sheet say about who created the record and what the reviewer's signature means? 5) What happens when the reviewer, who also has the analyst role, wants to approve a record of their own?

## Using NCA Assistant in a Regulated Environment

This chapter is for users who report results from the app in a clinical study, a regulatory submission or another setting under a quality system. It does not replace your organisation's procedures for computerised systems.

### Status and Scope

NCA Assistant is open-source software without warranty or supplier support, and it is not validated for your use. It performs average bioequivalence only, not ABEL or RSABE. The public instance on shinyapps.io is outside the scope of any qualification: its package versions are set by the deployment, not by the validated release. For records under 21 CFR Part 11 or EU GMP Annex 11, use a controlled installation; controlled mode is qualified for the server on which it was qualified, not in general.

### Which Rules Apply

Bioequivalence and drug interaction studies are clinical trials, so good clinical practice applies: ICH E6(R3) and the EMA guideline on computerised systems and electronic data in clinical trials (EMA/INS/GCP/112288/2023). EU GMP Annex 11 and 21 CFR Part 11 apply where your organisation or the use falls under them. Your own procedure for validating computerised systems decides what is needed; the validation package gives the evidence to start from.

### Installing a Qualified Instance

Download a tagged release ([github.com/Robterheine/NCAassistant/releases](https://github.com/Robterheine/NCAassistant/releases)), install it on a computer or server whose access you control, and install the validated package versions with `Rscript install_and_run.R --validated` (see Installing the App on Your Own Computer). The plain `install_and_run.R` installs the latest packages from CRAN, which is fine for evaluation but is not the qualified configuration.

### Qualification

The validation folder holds the User Requirement Specification, the IQ/OQ/PQ protocol and the test script. In outline:

1. Review the URS and its risk assessment against your intended use, and approve the protocol before execution (protocol section 1.2).
2. Run `Rscript validation/validation.R` and keep `validation_results.csv` and `validation_environment.txt`. The installation checks IQ-REL-01 and IQ-REL-02 compare your files and package versions with the release.
3. Carry out the manual OQ tests (protocol section 4) and the user PQ (section 5): analyse datasets like your own studies and compare the results with an independent reference.
4. Assess every deviation and every SUPPORTIVE failure, and have QA sign the outcome (sections 7 and 8).

The checklist for adopting organisations in section 1.3 of the protocol lists these steps.

## Controls on a Controlled Installation, and Your Responsibilities

On the public app and in a standard installation, the app has no user accounts, no audit trail and no electronic signatures. Put these controls in place around it:

- access to the qualified installation restricted to trained users;
- a log entry for every analysis: who ran it, when, on which data file (with its SHA-256), with which release, and for what purpose;
- a second person who checks the settings against the protocol and the results before they are used;
- archiving of the source files and the Analysis Records in a system with an audit trail.

A controlled installation provides personal accounts with roles, a password policy, an audit trail, stored records, review signatures, an audit trail page and archives (Working on a Controlled Installation, and Appendix E). What stays with your organisation:

- the server as set up in Appendix E, reached only through two-factor sign-in from managed devices, with a monitored system log;
- a named system owner who is not an analyst or reviewer on the same studies, and root access only for IT;
- account management: an identity check before an account is given out, accounts created or reset only when the person signs in at once, reporting when admin does not work, a periodic review of accounts, and deactivation on the day someone leaves;
- training records, and a written policy that electronic signatures are binding (for submissions to the FDA, also the certification letter of 21 CFR 11.100(c));
- review of the audit trail at a fixed interval and before each study report, with extra reviews after failed sign-ins, a broken chain or repeated runs, and filing the trail head in your quality system;
- an investigation, with corrective action, of a broken chain or an unexplained gap between an app stop and start;
- producing the trail in an inspection: the CSV export of the study, the signature sheets, and on request the archive with its verification report, and an inspector account for the inspection;
- archives at the end of each study and when the server is retired, backups at a separate site, restore tests and a disaster recovery plan;
- a privacy notice for users, and patching, security testing and incident response for the server.

**Limits you accept.** A new account can be taken over with the starting password admin by someone who knows the user ID, until its owner signs in: create accounts only when the owner signs in at once. The audit trail shows any change, but someone with root access to the server could rebuild it; the heads you file, the signature sheets and the backups are what show this. Reviewers see every study on the server: separate sponsors need separate installations. A deactivated user's open session runs on until the 15-minute timeout, although signing is refused at once.

## Before Each Analysis

On a controlled installation, check that the top bar shows the shield icon and your name. Check on the About page that the version and the pipeline SHA-256 are those of your qualified release (the release manifest, `validation/release_manifest.csv`, lists the hash of every file). Record the SHA-256 of the source data file. Check the settings against the protocol and the statistical analysis plan: design, Reference treatment, model, acceptance limits and their scope, BLQ rule and LLOQ, partial AUC intervals and their roles.

## Exclusions, Outliers and Deviations

The app has no control to exclude a subject or a period, so an exclusion means preparing a second data file. ICH M13A (2.2.1) asks for the criteria to be defined in the protocol and for any exclusion to be documented before the bioanalysis. Extreme values should not be removed on statistical grounds alone; only contemporaneously documented protocol violations justify removal (M13A 2.2.1.1). A subject without measurable concentrations, or with very low ones (AUC below 5% of the geometric mean), may be excluded only as an exception, in general for no more than one subject. A period with a pre-dose concentration above 5% of C<sub>max</sub> is excluded from the primary analysis (2.2.3.3).

A procedure that keeps the chain from source to result intact:

1. Record the SHA-256 of the unedited source file.
2. Make the analysis file with a documented, repeatable step, for example a short R script under version control, rather than by hand.
3. Keep both files, the list of exclusions with the reason for each, and the protocol reference.
4. Run the analysis on the analysis file and create the Analysis Record.
5. File the source hash and the exclusion list with the record.
6. Where the plan asks for it, run the analysis with and without the excluded data and report both.

## Reading Verdicts in a Regulated Context

- A verdict is computed on all the data you uploaded. When the checks name data that ICH M13A excludes, the verdict is not the M13A primary analysis (Path 6, Running and Reading the Results).
- Widened limits are not ABEL or RSABE, and All compared metrics also widens AUC to last point and AUC to infinity, which no guideline allows for bioequivalence. To widen a partial AUC, choose C<sub>max</sub> and partial AUCs.
- ICH M13A allows a fixed or a mixed model, but its primary analysis includes only subjects with data for both treatments (2.2.3.2). The mixed model also uses subjects with one treatment, which is not that population; use it only when the plan specifies it.
- For parallel groups the pooled-variance interval is the app's primary result; if the plan specifies an unequal-variance (Welch) analysis, that is your primary result.

## The Analysis Record in a Regulated Context

The record documents one analysis and shows that it can be reproduced with the same code and packages (Complete Analysis Record, in the chapter on using the app). It does not re-derive the bioequivalence statistics, it is not signed, and whoever changes a file can recompute its hash. Store the zip, or its SHA-256, in a controlled system. The Analyst name and Study name fields are typed, not signed in: record who ran the analysis in your log. For a manual selection of half-life points, the record keeps the final selection but not the reason: write the reason in the study file. On a controlled installation the record is stored on the server, the analyst is the signed-in user, and the review signature is kept in the audit trail and shown on the signature sheet, bound to the record's SHA-256.

## Reporting and Citing

Cite the software with its version: ter Heine R. NCA Assistant (v1.8.0). Radboud Applied Pharmacometrics, Radboudumc, Nijmegen, The Netherlands.  
<https://github.com/Robterheine/NCAassistant>. Add the pipeline SHA-256, the reference of your qualification, and the SHA-256 of the Analysis Record. The Example Methods Paragraphs on the Statistical Methods page describe the calculations; they say the analysis follows ICH M13A, EMA and FDA guidance, so check that your study design and analysis do before you keep that sentence. For a sample-size justification, record the planner's inputs and result (Path 1).

## Archiving and Retention

File the source data file, the analysis file, the Analysis Record, the SHA-256 of the zip, the release tag, validation/renv.lock and the qualification reference together. Keep them as long as your procedures and the law require; in the EU the trial master file of a clinical trial is kept for at least 25 years (Regulation (EU) No 536/2014, Article 58). The app keeps nothing: download the record before you close the session. On a controlled installation, the system owner archives the audit trail and the records with `manage_users.R` archive (Appendix E), which also keeps the software needed to verify and restore them.

## Upgrades and Change Control

Read the version history on the About page before you upgrade. Entries marked "Results can differ" and "Correctness fix" tell you which analyses a new version may change. A new release, a new R version or a new package version needs at least a targeted re-qualification (URS section 7); decide in advance whether completed analyses are repeated. On a controlled installation the audit trail continues across versions; each app start records the version and configuration in use.

## Data Protection

Use a local installation for real study data. Pseudonymised data are still personal data under the GDPR (Article 4(5)), and pharmacokinetic data from a trial are health data (Article 9). Sending them to a hosted service needs a processor agreement and, for a host outside the EU, a transfer basis. Upload only synthetic, example or anonymised data to the public instance. The audit trail of a controlled installation holds personal data about its users: who did what, and when. Tell users in a privacy notice, and keep the trail as long as the records it belongs to.

## Reporting Problems

Report problems at [github.com/Robterheine/NCAassistant/issues](https://github.com/Robterheine/NCAassistant/issues), with a description of the data (never the data themselves, unless synthetic). Correctness fixes are listed in the version history under "Correctness fix", with the situations they affect.

# Introduction to Non-Compartmental Analysis

## What Is Non-Compartmental Analysis?

Non-compartmental analysis (NCA) is a method for describing the pharmacokinetics of a drug without assuming a specific structural model for how the drug distributes in the body. Instead of fitting a mathematical model, NCA calculates pharmacokinetic parameters directly from the observed concentration–time data using numerical integration and regression.

NCA is the standard approach for pharmacokinetic data in drug development. EMA, FDA and PMDA use it for bioequivalence studies, and it is routine in Phase I trials and dose-finding studies.

## The Concentration–Time Curve

After a drug is administered, blood samples are collected at specified times and the drug concentration is measured. When plotted against time, the curve shows absorption (concentrations rising), distribution (transition), and elimination (concentrations declining). NCA extracts quantitative parameters from this curve without assuming its shape.

## Key Pharmacokinetic Parameters

| Parameter                 | Symbol                        | What it tells you                                                                                                      |
|---------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Peak concentration        | C <sub>max</sub>              | The highest concentration observed. Relevant for efficacy and safety.                                                  |
| Time of peak              | T <sub>max</sub>              | When the peak occurs. Reflects the rate of absorption.                                                                 |
| AUC to last point         | AUC <sub>0–t</sub>            | Exposure up to the last measurable concentration, from the observed data only. The primary AUC in bioequivalence.      |
| AUC to infinity           | AUC <sub>0–∞</sub>            | AUC <sub>0–t</sub> plus the extrapolated tail, C <sub>last</sub> /λ <sub>z</sub> . Total exposure after a single dose. |
| AUC % extrapolated        | %AUC <sub>extrap</sub>        | The share of AUC <sub>0–∞</sub> that is extrapolated. Above 20% the value rests largely on the extrapolation.          |
| Elimination rate constant | λ <sub>z</sub>                | The slope of the terminal phase on a log scale, from a regression on its last points.                                  |
| Adjusted R <sup>2</sup>   | R <sup>2</sup> <sub>adj</sub> | How well the terminal regression fits; the app leaves λ <sub>z</sub> -based parameters empty below a minimum.          |
| Terminal half-life        | t <sub>1/2</sub>              | ln 2/λ <sub>z</sub> : the time for the concentration to halve during the terminal phase.                               |

|                                 |                                    |                                                                                                                                                                      |
|---------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apparent clearance              | CL/F                               | Clearance divided by the bioavailability F: the volume of plasma cleared of drug per unit time.                                                                      |
| Apparent volume of distribution | V <sub>z</sub> /F                  | The terminal volume of distribution divided by F.                                                                                                                    |
| Mean residence time             | MRT                                | The average time a drug molecule stays in the body; after an oral dose it includes the absorption time.                                                              |
| Partial AUC                     | AUC t <sub>1</sub> –t <sub>2</sub> | Exposure over part of the profile, for example the first 30 minutes. It catches a difference in the shape of the curve that C <sub>max</sub> and total AUC can miss. |

Appendix A lists every parameter the app reports, with its unit.

## Why NCA Is Used So Widely

NCA does not assume a compartmental structure, so the same calculations apply whether a drug follows one-compartment kinetics or multi-phasic elimination. Results depend on the observed data and the numerical method, not on model selection, which makes them reproducible across software and analysts.

Regulatory agencies accept NCA because the methods are well-defined and transparent. NCA is the standard approach for bioequivalence with pharmacokinetic endpoints, where C<sub>max</sub> (rate) and AUC (extent) are the primary endpoints.

## When to Use NCA vs. Compartmental Modelling

Use NCA when you need standard PK parameters from observed data: first-in-human studies, bioequivalence, drug interaction studies, dose proportionality, food effect studies. Use compartmental modelling when you need to simulate unobserved scenarios, when you have sparse sampling or irregular dosing, or when you need to understand mechanisms. The two approaches are complementary.

## Theoretical Background

### Trapezoidal Methods for AUC Calculation

#### The Linear Trapezoidal Rule

Between consecutive observations at  $t_i$  and  $t_{i+1}$  with concentrations  $C_i$  and  $C_{i+1}$ , the area is approximated as a trapezoid:

$$\text{AUC}(i) = (t_{i+1} - t_i) \times (C_i + C_{i+1}) / 2$$

This assumes linear concentration change. During elimination, drug concentrations typically decline exponentially, and the linear assumption leads to systematic overestimation: a straight line between two points on an exponential decay lies above the true curve.

#### The Logarithmic Trapezoidal Rule

If the concentration declines mono-exponentially between two points, the exact integral yields:

$$\text{AUC}(i) = (t_{i+1} - t_i) \times (C_i - C_{i+1}) / (\ln C_i - \ln C_{i+1})$$

This gives a more accurate estimate during the terminal phase. It requires both concentrations to be positive.

#### The Linear-Up / Log-Down Hybrid

Proposed by Yeh and Kwan (1978): the linear formula while concentrations rise and the logarithmic formula while they fall. It is the default in NCA Assistant. Choose the method in the protocol: ICH M13A (2.2.2.2) asks for it to be reported, and the two methods give slightly different results (in Tutorial 4, an AUC to last point ratio of 111.25% with linear-up/log-down and 111.16% with the linear method).

#### Two Special Cases

A fall to zero. The logarithmic formula needs two positive concentrations, so it cannot be used when a BLQ sample set to 0 follows a measurable one. NonCompart then adds no area for that segment. The app uses the linear formula there instead, so the fall to zero is counted in AUC to last point,  $\text{AUC}_\infty$  and everything derived from them.

An IV bolus with a sample at time 0. After an IV bolus the concentration at time 0 is  $C_0$ , the value just after the injection, not a pre-dose value. The app sets aside samples at or before time 0 and back-extrapolates  $C_0$  log-linearly from the first two concentrations when they decline; otherwise  $C_0$  is the first measured concentration.

### Terminal Phase Characterisation and Half-Life Estimation

#### What Is the Terminal Phase?

After distribution is complete, the concentration declines as  $C(t) = A \cdot e^{-\lambda_z t}$ , where  $A$  is the intercept of the terminal phase. On a log-linear plot, this appears as a straight line with slope  $-\lambda_z$ . The half-life is  $t_{1/2} = \ln(2) / \lambda_z$ .

#### Automated Point Selection

NCA Assistant uses NonCompart's slope selection, the same for the results table and the Half-Life Review. It fits log-linear regressions to the last 3, 4, 5, ... points (after extravascular dosing only points after C<sub>max</sub> are used, and never values set by a BLQ rule) and takes the fit with the highest adjusted R<sup>2</sup>; fits within 0.0001 of the best prefer more points. If that fit has an adjusted R<sup>2</sup> below the minimum (slider, default 0.70), the profile gets no half-life, and AUC<sub>∞</sub>, CL/F, Vz/F and MRT are left empty too; a note names the profile. This happens when the terminal phase is poorly sampled, when late samples are near the LLOQ, or when elimination is multi-phasic. A lower minimum accepts noisier fits. Judging the Terminal Phase, in Reading Your Results, lists what to check before you report a half-life.

### Manual Point Selection

Adjust the points yourself for multi-phasic elimination, enterohepatic recirculation or late samples near the LLOQ. In Half-Life Review, select a profile, tick the terminal-phase points and click Recalculate. The slope must be negative, and although 2 points are accepted, a fit to be reported needs at least 3. NonCompart then recomputes half-life, AUC<sub>∞</sub>, CL/F, Vz/F and MRT from those points, with its unit conversions. A hand-picked selection is used whatever its adjusted R<sup>2</sup>, so judge the fit on the plot. Selections are kept when you run the analysis again and are listed in the Analysis Record.

With flip-flop kinetics (absorption slower than elimination, as with many modified-release products) the terminal slope reflects absorption. No choice of points corrects that: report the value as an apparent half-life, and do not derive elimination from it. Only IV data can tell the two apart.

### Derived Parameters

With λ<sub>z</sub> and the observed last concentration C<sub>last</sub>, NCA Assistant (through NonCompart) calculates:

$AUC_{0-\infty} = AUC_{0-t} + C_{last} / \lambda_z$  (observed); with the predicted last concentration from the regression, the "predicted" variants

$AUC \% \text{ extrapolated} = (AUC_{0-\infty} - AUC_{0-t}) / AUC_{0-\infty} \times 100$

$CL/F = Dose / AUC_{0-\infty}$  and  $V_z/F = Dose / (\lambda_z \times AUC_{0-\infty})$ , without /F after an IV dose

$MRT = AUMC_{0-\infty} / AUC_{0-\infty}$ ; after an infusion, minus half the infusion duration

The lag time T<sub>lag</sub> is the last sampling time before the first measured concentration. After an IV bolus, AUC % back-extrapolated is the share of AUC between time 0 and the first sample, from the back-extrapolated C<sub>0</sub>. The Statistical Methods page in the app gives the same formulas.

### BLQ Handling: Pharmacological Rationale

Every assay has a lower limit of quantification (LLOQ). How BLQ samples are handled changes AUC. Setting early BLQ samples to 0 is usual before absorption; treating late BLQ samples as missing avoids adding area that was never measured. LLOQ/2 is a compromise that adds a little area wherever it is used. Rule 1, the default, follows the usual convention and ICH M13A, which treats BLQ values as zero. Rules 4 and 6 put LLOQ/2 into the absorption phase: that adds area during a lag, though the lag time itself is not changed, because a value set by a rule is not a measured concentration. They belong in sensitivity analyses, or in a protocol that asks for them. Running two rules and comparing the results

shows how much the choice matters. Beal (2001) compares methods for BLQ data in model-based (population) analysis, not in NCA.

## Steady-State Considerations

At steady state  $AUC_{\tau}$  replaces  $AUC_{\infty}$ . You enter the dosing interval  $\tau$ ;  $AUC_{\tau}$  is the AUC from 0 to  $\tau$ , interpolated between samples and extrapolated with  $\lambda_z$  when the last sample is missing or earlier than  $\tau$  (not reported if that fit was rejected).  $C_{avg} = AUC_{\tau}/\tau$ , fluctuation =  $(C_{max} - C_{min})/C_{avg} \times 100$  and swing =  $(C_{max} - C_{min})/C_{min}$ .  $C_{min}$  is the lowest observed concentration from 0 to  $\tau$ , the pre-dose sample included, and  $C_{\tau}$  is the observed concentration at  $\tau$ ; the two differ when the minimum falls after the dose.  $CL/F$  and  $V_z/F$  are calculated from  $AUC_{\tau}$ .  $AUC_{\infty}$ , its % extrapolated,  $AUMC_{\infty}$ , and the single-dose  $V_{ss}$  and  $MRT$  are not meaningful during repeated dosing and are left empty. A half-life from one interval is unreliable when the interval does not cover the terminal phase.

## Partial AUCs: Interpolation and the End of the Interval

A partial AUC is the area over part of the profile. With both cutoffs on sampling times it is simply the sum of the trapezoids between them.

A cutoff between two samples has to be interpolated: linearly, or log-linearly on a falling segment when you chose linear-up/log-down. The two methods differ most where the curve bends, which is why a protocol that asks for a partial AUC should schedule a sample at the cutoff.

An interval can end at  $t$ , the time of the last measurable concentration. That time belongs to the profile, not to the sampling schedule: a later sample that came back below the limit of quantification does not extend it. The app calculates such an interval as  $AUC_{0-t}$  minus the AUC from 0 to the start, so an interval ending exactly at the last measurable time gives the same number.

Past that point nothing is reported. Extrapolating with  $\lambda_z$  would turn an observed exposure metric into a model-based one, and it would do so for the profiles that carry the least information.

## Statistical Methods

### The Two One-Sided Tests (TOST) Procedure

Conventional hypothesis testing ( $H_0: \mu_T = \mu_R$ ) can only reject equality, not prove similarity. Schuirmann (1987) reformulated the problem as two one-sided tests:

$H_{01}: \mu(T)/\mu(R) \leq 0.80$  (ratio too low)

$H_{02}: \mu(T)/\mu(R) \geq 1.25$  (ratio too high)

Both rejected at  $\alpha = 0.05 \rightarrow$  ratio lies within  $[0.80, 1.25] \rightarrow$  bioequivalent. This is equivalent to: the 90% CI for the ratio falls entirely within  $[80\%, 125\%]$ . The limits are symmetric on the log scale:  $\ln(0.80) = -0.2231$ ,  $\ln(1.25) = +0.2231$ .

### The ANOVA Model for Crossover Designs

The standard model for a  $2 \times 2 \times 2$  crossover:

$$\ln(Y) = \mu + \text{SEQ} + \text{SUBJ}(\text{SEQ}) + \text{PER} + \text{TRT} + \varepsilon$$

SEQ accounts for differences between sequence groups, SUBJ removes between-subject variability from the residual, PER accounts for changes between periods, and TRT is the effect of interest: its coefficient estimates the log-ratio of geometric means. NCA Assistant fits this model for all crossover designs, with Sequence, Subject, Period and Treatment as factors; with fixed effects this is EMA Method A.

### Fixed-Effects vs. Mixed-Effects Models

Fixed effects (lm): all terms, including Subject, are fixed (EMA Method A). Subjects without both treatments do not contribute to the comparison. Residual df =  $N - 2$  for a balanced  $2 \times 2 \times 2$ .

Mixed effects (nlme::lme): Subject is a random intercept, and denominator df use the containment method. Subjects with only one treatment still contribute. ICH M13A (2.2.3.2) allows a general linear or a mixed model, but its primary analysis includes all data of the subjects with evaluable data for both products; with the mixed model the app also uses subjects with one treatment, which is not that population. Use the model your protocol or statistical analysis plan specifies, for example the mixed model as a sensitivity analysis. If the mixed model was chosen and cannot be fitted, the app shows the fixed-effects result for information, says so in the Model column, and gives no verdict: the model is not changed after the data are seen. This is EMA Method B (one within-subject variance; results agree with replicateBE). It is not the FDA model for replicate designs, which has separate Test and Reference variances, a subject-by-formulation interaction and Satterthwaite degrees of freedom.

### Confidence Interval Construction

$$90\% \text{ CI} = \exp(\delta \pm t(0.95, \text{df}) \times \text{SE}(\delta)) \times 100\%$$

where  $\delta$  is the treatment coefficient on the log scale, SE its standard error from the model, and df the denominator degrees of freedom. The limits are rounded to two decimals before comparison with the acceptance limits, as the EMA guideline (2010, section 4.1.8) and the

FDA (2026) ask. Bioequivalence is concluded when the rounded interval lies within the limits (80.00–125.00% by default) and, for wider limits, the point estimate lies within 80.00–125.00%. A verdict is given only for the 90% interval, which corresponds to the two one-sided tests at  $\alpha = 0.05$ ; at another level the app shows the interval without a verdict. Widened limits apply to Cmax only unless you choose otherwise (Path 6, Widened and Narrowed Limits).

## Design-Specific Models

**Paired comparison:**  $\ln(Y) = \mu + \text{SUBJ} + \text{TRT} + \varepsilon$ , equivalent to a paired t-test;  $df = N - 1$ . Ratio and interval only, no verdict.

**Parallel design:**  $\ln(Y) = \mu + \text{TRT} + \varepsilon$ .  $df = N_T + N_R - 2$ . Differences between subjects enter the comparison, so more subjects are needed than in a crossover. The interval assumes equal variances in both groups (pooled variance). The app also calculates the Welch interval, which allows unequal variances, and reports it as a sensitivity result when it leads to a different conclusion. When the analysis plan specifies an unequal-variance analysis, that is the primary result, not the app's verdict.

**Replicate designs:** the same model, with Period at 3 or 4 levels; each administration enters once. The confidence interval uses average bioequivalence limits. The within-subject variability table uses a separate model per treatment (Sequence, Subject and Period, fitted to subjects with that treatment at least twice), as in the replicateBE package; the app's results agree with replicateBE on its 30 reference data sets.

**Tmax** is compared as an untransformed difference with a t-based interval, without a verdict. Because Tmax takes only the sampling times, it is usually summarised as median and range and compared nonparametrically, outside the app.

## Drug–Drug Interaction Analysis

DDI studies use the same statistics as BE: the ANOVA model, the geometric mean ratio and its 90% confidence interval. The difference is the interpretation: the ratio is the fold-change in exposure caused by the interacting drug, judged against no-effect boundaries for the victim drug (see Analysing Drug–Drug Interaction Studies) rather than BE acceptance limits.

ICH M12 on drug interaction studies (2024), published as guidance by EMA and FDA, asks for the geometric mean ratio of AUC and Cmax (and Cmin when relevant) with and without the precipitant, with its 90% confidence interval (section 5.2). Crossover studies, including one-sequence studies, are preferred over parallel designs (section 3.2.1).

## Power and Sample Size for Bioequivalence

Exact power via Owen's Q function for standard ABE. For ABEL, RSABE and NTID, power is simulated in PowerTOST: 100,000 simulated studies for the sample size, 10,000 for power and the power curve. Inputs: within-subject CV (for scaled methods, of Test and Reference), expected ratio, target power and design. A higher CV or a ratio further from 100% needs more subjects; a crossover needs fewer subjects than a parallel design.

Practical guidance: without data, a CV of 30% is a cautious assumption. The expected ratio defaults to 95% for standard bioequivalence, 90% for highly variable drugs and 97.5% for

narrow therapeutic index drugs; target power 80%, or 90% for studies where failure is costly. Check the Sample Size vs. Variability tab to see how N changes with the CV.

## Partial AUCs in Bioequivalence

Regulators ask for a partial AUC when two products can share C<sub>max</sub> and total AUC while the shape of the curve differs. FDA product-specific guidances name the metric and the interval per product; Hopefl et al. (2025) list the guidances published between 2020 and 2024.

The statistics do not change. A partial AUC is log-transformed and analysed with the same model and confidence interval as C<sub>max</sub> and AUC. The interval comes from the protocol; choosing it after seeing the data inflates the chance of a false conclusion of equivalence.

A pivotal metric has to meet the acceptance limits, a supportive one is reported without a verdict. With several pivotal metrics each has to pass on its own. That is an intersection-union test, so no correction for multiple testing applies.

Zeros need care. An early interval with only BLQ samples can be exactly 0, which has no logarithm. Leaving those profiles out would remove the lowest exposures and push the ratio up, so the app reports no estimate and no verdict for that metric and shows how many zero values each treatment had.

For every metric the table also counts, per treatment, the profiles that had no value, the profiles with a value of zero, and the profiles resting mainly on concentrations set by the BLQ rule. The last count matters because such a profile can pull a ratio on the log scale without ever being zero: under Rule 1 an early window may be exactly zero and get no verdict, while under Rule 6 the same sample becomes LLOQ/2 and the metric keeps a verdict that one imputed value helped to decide.

The EMA allows the acceptance criteria for a partial AUC to be widened in the same way as for C<sub>max</sub> of a highly variable product (EMA/CHMP/EWP/280/96 Rev1, section 6.8.2.2). The app does not scale limits: for a replicate design it shows CV<sub>wR</sub> and the limits it would imply, and it applies the limits you enter. To widen C<sub>max</sub> and the partial AUCs but not AUC to last point or to infinity, choose C<sub>max</sub> and partial AUCs under Widened limits apply to. The same guideline recommends an early and a terminal partial AUC, split at a pre-specified cut-off such as half the dosing interval, when a single-dose study replaces a multiple-dose study.

Plan such a study with the CV of the partial AUC itself. For loxapine inhalation powder the within-subject CV was above 30% for AUC 0–10 min but below 30% for AUC 0–30 min, which is why the 30-minute window became the pivotal metric (Hopefl et al., 2025).

## Regulatory Guidelines

**Which guideline applies.** For immediate-release solid oral dosage forms, ICH M13A (in effect in the EU since January 2025) is the bioequivalence guideline. The EMA guideline of 2010 still applies to other dosage forms and to the widening of limits for highly variable drugs, which M13A does not cover; later parts of the ICH M13 series address such topics. In the USA, FDA guidances and product-specific guidances apply alongside M13A. Check the current status of each document before you cite it.

**ICH M13A Bioequivalence for Immediate-Release Solid Oral Dosage Forms** (Step 4, 23 July 2024; in effect in the EU from 25 January 2025; published by ICH, EMA and FDA)

**EMA Guideline on the Investigation of Bioequivalence** (CPMP/EWP/QWP/1401/98 Rev. 1, 2010): <https://www.ema.europa.eu/en/investigation-bioequivalence-scientific-guideline>

**EMA product-specific bioequivalence guidance**, and the EMA questions and answers on bioequivalence, which describe Methods A and B: on the EMA website

**FDA Statistical Approaches to Establishing Bioequivalence** (May 2026; replaces the 2001 guidance): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/statistical-approaches-establishing-bioequivalence>

**FDA Bioequivalence Studies With Pharmacokinetic Endpoints for Drugs Submitted Under an ANDA**, and **FDA product-specific guidances** for generic drugs, on the FDA website; **21 CFR Part 320** (bioavailability and bioequivalence requirements)

**FDA Bioavailability Studies Submitted in NDAs or INDs: General Considerations** (draft, 2022): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-studies-submitted-ndas-or-inds-general-considerations>

**ICH M12 Drug Interaction Studies** (2024; published by EMA and FDA): <https://www.ema.europa.eu/en/ich-m12-drug-interaction-studies-scientific-guideline>

**EMA Guideline on the pharmacokinetic and clinical evaluation of modified release dosage forms** (EMA/CHMP/EWP/280/96 Rev1, 2014)

**ICH E6(R3) Good Clinical Practice** (2025), and the **EMA Guideline on computerised systems and electronic data in clinical trials** (EMA/INS/GCP/112288/2023)

**CDISC SDTM Controlled Terminology** (release 2026-03-27, source of the parameter codes): <https://evs.nci.nih.gov/ftp1/CDISC/SDTM/>

## Appendices

### Appendix A: Output Parameters

Every column the NCA can produce, with its label in the app, its definition and its unit for a dose in mg, time in h and concentrations in ng/mL. The unit follows the units you select. "With  $\lambda_z$ " means the parameter needs an accepted terminal fit; "single dose" means it is left empty at steady state. Partial AUC columns (Partial AUC t1–t2, Cmax t1–t2, Tmax t1–t2) have the unit of AUC, concentration and time. CL/F, Vz/F and the bioequivalence comparison of AUC to infinity use the observed AUC0– $\infty$ .

| Code                | Label in the app                       | Definition                                                 | Unit (example) | Reported for          |
|---------------------|----------------------------------------|------------------------------------------------------------|----------------|-----------------------|
| b0                  | Y-Intercept (ln scale)                 | Intercept of the terminal regression on the ln scale       | (none)         | All, with $\lambda_z$ |
| C <sub>MAX</sub>    | Peak Concentration (C <sub>max</sub> ) | Highest observed concentration                             | ng/mL          | All                   |
| T <sub>MAX</sub>    | Time of Peak (T <sub>max</sub> )       | Time of C <sub>max</sub>                                   | h              | All                   |
| T <sub>LAG</sub>    | Lag Time                               | Last sampling time before the first measured concentration | h              | Extravascular         |
| CL <sub>ST</sub>    | Last Measurable Concentration          | Last measurable concentration (C <sub>last</sub> )         | ng/mL          | All                   |
| CL <sub>STP</sub>   | Predicted Last Concentration           | C <sub>last</sub> predicted by the terminal regression     | ng/mL          | All, with $\lambda_z$ |
| T <sub>LST</sub>    | Time of Last Measurable Conc           | Time of C <sub>last</sub> (t)                              | h              | All                   |
| LAM <sub>ZHL</sub>  | Half-Life                              | Terminal half-life, $\ln 2/\lambda_z$                      | h              | All, with $\lambda_z$ |
| LAM <sub>Z</sub>    | Elimination Rate Constant              | Terminal elimination rate constant $\lambda_z$             | 1/h            | All, with $\lambda_z$ |
| LAM <sub>ZLL</sub>  | Lambda_z Lower Time                    | First time point of the terminal regression                | h              | All, with $\lambda_z$ |
| LAM <sub>ZUL</sub>  | Lambda_z Upper Time                    | Last time point of the terminal regression                 | h              | All, with $\lambda_z$ |
| LAM <sub>ZNPT</sub> | Points Used for Half-Life              | Number of points in the terminal regression                | (none)         | All, with $\lambda_z$ |
| CORRXY              | Correlation (r)                        | Correlation coefficient of the terminal regression         | (none)         | All, with $\lambda_z$ |

|         |                                |                                                                                       |                      |                                 |
|---------|--------------------------------|---------------------------------------------------------------------------------------|----------------------|---------------------------------|
| R2      | R-squared                      | R <sup>2</sup> of the terminal regression                                             | (none)               | All, with $\lambda_z$           |
| R2ADJ   | Adjusted R-squared             | Adjusted R <sup>2</sup> of the terminal regression; the minimum applies to it         | (none)               | All, with $\lambda_z$           |
| AUCLST  | AUC to Last Point              | AUC from 0 to Tlast (AUC0–t)                                                          | ng/mL·h              | All                             |
| AUCALL  | AUC All (incl. trailing zero)  | AUC from 0 to the last sample, including a trailing zero                              | ng/mL·h              | All                             |
| AUCIFO  | AUC to Infinity (observed)     | AUC0–t + Clast/ $\lambda_z$                                                           | ng/mL·h              | Single dose, with $\lambda_z$   |
| AUCIFP  | AUC to Infinity (predicted)    | AUC0–t + Clast,pred/ $\lambda_z$                                                      | ng/mL·h              | Single dose, with $\lambda_z$   |
| AUCPEO  | AUC % Extrapolated (observed)  | Share of AUC0– $\infty$ (observed) that is extrapolated, %                            | (none)               | Single dose, with $\lambda_z$   |
| AUCPEP  | AUC % Extrapolated (predicted) | Share of AUC0– $\infty$ (predicted) that is extrapolated, %                           | (none)               | Single dose, with $\lambda_z$   |
| AUMCLST | AUMC to Last Point             | Area under the first-moment curve ( $C \times t$ ) from 0 to Tlast                    | ng/mL·h <sup>2</sup> | All                             |
| AUMCIFO | AUMC to Infinity (observed)    | AUMC extrapolated to infinity with Clast                                              | ng/mL·h <sup>2</sup> | Single dose, with $\lambda_z$   |
| AUMCIFP | AUMC to Infinity (predicted)   | AUMC extrapolated to infinity with Clast,pred                                         | ng/mL·h <sup>2</sup> | Single dose, with $\lambda_z$   |
| AUMCPEO | AUMC % Extrapolated (obs)      | Share of AUMC0– $\infty$ (observed) that is extrapolated, %                           | (none)               | Single dose, with $\lambda_z$   |
| AUMCPEP | AUMC % Extrapolated (pred)     | Share of AUMC0– $\infty$ (predicted) that is extrapolated, %                          | (none)               | Single dose, with $\lambda_z$   |
| VZFO    | Apparent Volume (Vz/F)         | Dose/( $\lambda_z \times \text{AUC0–}\infty$ ); at steady state from AUC <sub>T</sub> | L                    | Extravascular, with $\lambda_z$ |
| VZFP    | Apparent Volume (pred)         | As Vz/F, with the predicted AUC0– $\infty$                                            | L                    | Extravascular single dose       |
| CLFO    | Apparent Clearance (CL/F)      | Dose/AUC0– $\infty$ ; at steady state Dose/AUC <sub>T</sub>                           | L/h                  | Extravascular                   |

|            |                                           |                                                                                            |                             |                           |
|------------|-------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------|---------------------------|
| CLFP       | Apparent Clearance (pred)                 | As CL/F, with the predicted AUC <sub>0–∞</sub>                                             | L/h                         | Extravascular single dose |
| MRTEVLST   | Mean Residence Time (to last)             | AUMC <sub>0–t</sub> /AUC <sub>0–t</sub> , including absorption time                        | h                           | Extravascular single dose |
| MRTEVIFO   | Mean Residence Time (to inf, obs)         | AUMC <sub>0–∞</sub> /AUC <sub>0–∞</sub> (observed), including absorption time              | h                           | Extravascular single dose |
| MRTEVIFP   | Mean Residence Time (to inf, pred)        | As above, with the predicted values                                                        | h                           | Extravascular single dose |
| CMAX_DN    | Dose-Normalised C <sub>max</sub>          | C <sub>max</sub> divided by the dose of that profile                                       | ng/mL per mg                | With dose normalisation   |
| AUCLST_DN  | Dose-Normalised AUC Last                  | AUC <sub>0–t</sub> divided by the dose                                                     | ng/mL·h per mg              | With dose normalisation   |
| AUCIFO_DN  | Dose-Normalised AUC Inf                   | AUC <sub>0–∞</sub> (observed) divided by the dose                                          | ng/mL·h per mg              | With dose normalisation   |
| AUCIFP_DN  | Dose-Normalised AUC Inf (pred)            | AUC <sub>0–∞</sub> (predicted) divided by the dose                                         | ng/mL·h per mg              | With dose normalisation   |
| AUMCLST_DN | Dose-Normalised AUMC Last                 | AUMC <sub>0–t</sub> divided by the dose                                                    | ng/mL·h <sup>2</sup> per mg | With dose normalisation   |
| AUMCIFO_DN | Dose-Normalised AUMC Inf                  | AUMC <sub>0–∞</sub> (observed) divided by the dose                                         | ng/mL·h <sup>2</sup> per mg | With dose normalisation   |
| AUMCIFP_DN | Dose-Normalised AUMC Inf (pred)           | AUMC <sub>0–∞</sub> (predicted) divided by the dose                                        | ng/mL·h <sup>2</sup> per mg | With dose normalisation   |
| AUCTAU     | AUC Within Dosing Interval                | AUC from 0 to $\tau$ ; extrapolated with $\lambda_z$ when the last sample is before $\tau$ | ng/mL·h                     | Steady state              |
| TAU        | Dosing Interval (tau)                     | The dosing interval you entered                                                            | h                           | Steady state              |
| CAVG       | Average Concentration (C <sub>avg</sub> ) | AUC $\tau$ / $\tau$                                                                        | ng/mL                       | Steady state              |
| CMIN_SS    | Minimum Concentration (C <sub>min</sub> ) | Lowest observed concentration from 0 to $\tau$ , pre-dose included                         | ng/mL                       | Steady state              |

|          |                                          |                                                                                     |        |                      |
|----------|------------------------------------------|-------------------------------------------------------------------------------------|--------|----------------------|
| CTAU_SS  | Concentration at Tau (Ctau)              | Observed concentration at $\tau$ ; empty without a sample at $\tau$                 | ng/mL  | Steady state         |
| FLUCTP   | Peak-Trough Fluctuation (%)              | $(C_{\max} - C_{\min})/C_{\text{avg}} \times 100$                                   | (none) | Steady state         |
| SWING    | Swing $((C_{\max} - C_{\min})/C_{\min})$ | $(C_{\max} - C_{\min})/C_{\min}$                                                    | (none) | Steady state         |
| C0       | Initial Concentration (C0)               | Concentration at time 0, back-extrapolated from the first two samples               | ng/mL  | IV bolus             |
| AUCPBEO  | AUC % Back-Extrapolated (observed)       | Share of $AUC_{0-\infty}$ (observed) between 0 and the first sample, %              | (none) | IV bolus single dose |
| AUCPBEP  | AUC % Back-Extrapolated (predicted)      | As above, with the predicted $AUC_{0-\infty}$                                       | (none) | IV bolus single dose |
| VZO      | Volume of Distribution (Vz)              | $\text{Dose}/(\lambda_z \times AUC_{0-\infty})$ ; at steady state from $AUC_{\tau}$ | L      | IV, with $\lambda_z$ |
| VZP      | Volume of Distribution (pred)            | As Vz, with the predicted $AUC_{0-\infty}$                                          | L      | IV single dose       |
| CLO      | Clearance (CL)                           | $\text{Dose}/AUC_{0-\infty}$ ; at steady state $\text{Dose}/AUC_{\tau}$             | L/h    | IV                   |
| CLP      | Clearance (pred)                         | As CL, with the predicted $AUC_{0-\infty}$                                          | L/h    | IV single dose       |
| MRTIVLST | Mean Residence Time IV (to last)         | $AUMC_{0-t}/AUC_{0-t}$ (minus half the infusion time)                               | h      | IV single dose       |
| MRTIVIFO | Mean Residence Time IV (to inf, obs)     | $AUMC_{0-\infty}/AUC_{0-\infty}$ (observed; minus half the infusion time)           | h      | IV single dose       |
| MRTIVIFP | Mean Residence Time IV (to inf, pred)    | As above, with the predicted values                                                 | h      | IV single dose       |
| VSSO     | Volume at Steady State (Vss)             | Volume at steady state, $MRT \times CL$                                             | L      | IV single dose       |
| VSSP     | Volume at Steady State (pred)            | As Vss, with the predicted values                                                   | L      | IV single dose       |

## Appendix B: Glossary

| Term                                    | Definition                                                                                                                        |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ABE                                     | Average bioequivalence (80–125% limits by default)                                                                                |
| ABEL                                    | Average bioequivalence with expanding limits (EMA, for highly variable drugs)                                                     |
| Adjusted R <sup>2</sup>                 | Goodness of fit of the terminal regression, corrected for the number of points; the minimum in the app applies to it              |
| ADNCA                                   | CDISC ADaM dataset for non-compartmental analysis                                                                                 |
| ANOVA                                   | Analysis of variance                                                                                                              |
| AUC                                     | Area under the concentration–time curve                                                                                           |
| AUC % extrapolated                      | The share of AUC to infinity that lies beyond the last measurable concentration                                                   |
| AUC to infinity (AUC <sub>0–∞</sub> )   | AUC to last point plus C <sub>last</sub> /λ <sub>z</sub>                                                                          |
| AUC to last point (AUC <sub>0–t</sub> ) | AUC from 0 to the last measurable concentration                                                                                   |
| AUC <sub>τ</sub>                        | AUC within the dosing interval (steady state)                                                                                     |
| Audit trail                             | On a controlled installation: the record of who did what, when and on which data, kept on the server and protected against change |
| BLQ                                     | Below the limit of quantification                                                                                                 |
| CL/F                                    | Apparent clearance: clearance divided by the bioavailability F                                                                    |
| C <sub>max</sub>                        | Maximum observed concentration                                                                                                    |
| Controlled installation                 | NCA Assistant on your organisation's server in controlled mode, with sign-in, audit trail, stored records and review signatures   |
| Crossover                               | A design in which each subject receives the treatments in turn, in periods separated by a washout                                 |
| CV%                                     | Coefficient of variation                                                                                                          |
| CV <sub>wR</sub> / CV <sub>wT</sub>     | Within-subject coefficient of variation of the Reference / Test                                                                   |
| DDI                                     | Drug–drug interaction                                                                                                             |
| EMA Method A / B                        | The fixed-effects (A) and mixed-effects (B) ANOVA of the EMA questions and answers on bioequivalence                              |
| Extravascular                           | Given other than into a vein (oral, intramuscular, subcutaneous), so the drug is absorbed first                                   |
| Geometric mean                          | exp(mean of the ln values); the usual summary for AUC and C <sub>max</sub>                                                        |
| Geometric CV                            | $100 \cdot \sqrt{(\exp(\text{SD}^2 \text{ of the ln values}) - 1)}$                                                               |
| GMR                                     | Geometric mean ratio (Test/Reference)                                                                                             |

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| HVD                       | Highly variable drug (within-subject CV of the Reference above 30%)                                                         |
| ICH M13A                  | ICH guideline on bioequivalence for immediate-release solid oral dosage forms (2024)                                        |
| Inspector                 | Account that can see records, the audit trail and the user overview, but cannot sign or change anything                     |
| Intersection-union test   | A test that passes only if each of several tests passes; no multiplicity correction is needed                               |
| $\lambda_z$               | Terminal elimination rate constant                                                                                          |
| LLOQ                      | Lower limit of quantification                                                                                               |
| MRT                       | Mean residence time                                                                                                         |
| MSE                       | Mean squared error: the residual variance of the ANOVA model                                                                |
| NCA                       | Non-compartmental analysis                                                                                                  |
| NTID                      | Narrow therapeutic index drug                                                                                               |
| Owen's Q                  | The function PowerTOST uses for exact power of the two one-sided tests                                                      |
| Partial AUC               | AUC over part of the profile, for example AUC 0–30 min; the interval comes from the protocol or a product-specific guidance |
| PE                        | Point estimate (geometric mean ratio)                                                                                       |
| Period                    | One administration in a crossover, with its own profile                                                                     |
| Pivotal metric            | A bioequivalence metric that has to meet the acceptance limits; it receives a verdict                                       |
| Power                     | The probability that a study shows bioequivalence when the products are bioequivalent as assumed                            |
| Precipitant (perpetrator) | The drug that causes a drug interaction                                                                                     |
| Profile                   | The concentrations of one subject under one treatment in one period                                                         |
| Reference                 | The treatment in the denominator of the ratio, chosen by the user                                                           |
| Reviewer                  | Account that approves or rejects records made by others and signs reviews of the audit trail                                |
| RSABE                     | Reference-scaled average bioequivalence (FDA, for highly variable drugs)                                                    |
| Sequence                  | The order in which a subject receives the treatments, for example TR or RT                                                  |
| Signature sheet           | Page showing a record's signatures: printed names, UTC times, meanings and the audit entries                                |
| Supportive metric         | A bioequivalence metric reported with its ratio and confidence interval, without a verdict                                  |
| System owner              | The person who manages accounts and archives on the server                                                                  |

|                     |                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| t (last measurable) | The time of the last measurable concentration of a profile; an interval written as 4 h–t ends there |
| t <sub>1/2</sub>    | Terminal elimination half-life                                                                      |
| T <sub>max</sub>    | Time of the maximum observed concentration                                                          |
| TOST                | Two one-sided tests                                                                                 |
| Victim              | The drug whose pharmacokinetics a drug interaction changes                                          |
| V <sub>z</sub> /F   | Apparent terminal volume of distribution: the volume divided by F                                   |
| Washout             | The time between periods of a crossover, long enough for the drug to disappear                      |
| Welch interval      | A confidence interval for a parallel design that does not assume equal variances in the two groups  |

In this manual, "missing" and "excluded" for a BLQ value mean the same: the value is left out of the calculation.

## Appendix C: Recognised Column Names

| Meaning       | Recognised names (case-insensitive)                                                   |
|---------------|---------------------------------------------------------------------------------------|
| Subject       | Subject, SubjectID, USUBJID, ID, Pat, PatID, Proband, Teilnehmer                      |
| Time          | TAD (preferred over TIME), Time, Hours, Hour, Tpt, NTIM, APTS, Zeit, Tid              |
| Concentration | Conc, Concentration, DV, Cp, Result, Konz, Plasma, or a name containing ng/mL or ug/L |
| Treatment     | Treatment, Trt, Formulation, Drug, Arm, Behandlung                                    |
| Period        | Period, Per, Prd, Phase, APERIOD, Occasion, OCC (not Visit)                           |
| Sequence      | Sequence, Seq, Grp                                                                    |
| Dose          | Dose, Amt, Amount, Dosis                                                              |

When no name matches, the app falls back to a column by position, so always check the suggested mapping. When the file has another column that could be the time or the concentration, a note says which one was chosen and names the other.

## Appendix D: Troubleshooting

### Opening and Installing

**“Disconnected from the server”:** the public app was idle too long, the analysis exceeded memory or time limits, or the app met an unexpected error. Reload the page, upload the file again and click Process Data; results that were not downloaded are lost. If it happens again, run the app locally and check the console.

**The installation stops with an error:** see Installing the App on Your Own Computer.

## Uploading and Processing

**Comma decimals not recognised:** set Decimal point to comma (and the Delimiter, often semicolon) in the upload settings.

**A decimal point in a decimal-comma file:** the file mixes decimal marks. Correct the values, or set the decimal mark to period if the file uses points throughout.

**“N BLQ entries detected but LLOQ is not set”:** set the LLOQ and click Process Data again; for “<X” entries, the button Set LLOQ to X fills in the value (Tutorial 2).

**“Profile(s) do not start near time zero”:** time is probably measured from the first dose of the study. Use the time since the dose of each period (in CDISC terms ARRLT or NRRLT, not AFRLT).

**Subject IDs that restart in each sequence:** give every subject an ID that is unique in the study, for example TR-01 and RT-01.

**“The data give concentrations in ...”:** the unit column of the file says one unit and the analysis settings another. Select the unit the file states, or correct the file.

**Large sampling gap warning:** for studies longer than 48 h the threshold is 48 h, so daily trough samples are not flagged; for shorter studies it is 24 h. A gap close to the threshold is usually normal sampling.

**“Refused: ...”** in the CDISC ADNCA upload: the message says why (for example derived records, several units, AFRLT as time, or AVAL missing without a BLQ result). Fix the dataset upstream or make the choice the message asks for.

## NCA

**Unit error before the NCA runs:** the combination of dose, time and concentration units is not supported, or molar units are combined with a mass dose without a molecular weight. Select supported units and enter the molecular weight.

**“NCA failed”:** the message gives the reason, usually no profile with at least 2 positive concentrations, or a mapping problem. Check the mapping and the quality report.

**Half-life is empty:** the automatic fit is below the minimum adjusted  $R^2$ , or fewer than 3 usable points remain. Review the profile in Half-Life Review, or lower the minimum if your analysis plan allows it.

**Half-life differs from other NCA software:** check the minimum  $R^2$ , the point selection in Half-Life Review, the BLQ rule and the trapezoidal method; programs differ in how they choose the terminal-phase points.

**My partial AUC is empty:** the interval reaches past the last measurable concentration of that profile, or starts after it. The note above the results names the profile and its last measurable time. Partial AUCs are not extrapolated.

## Bioequivalence

**“Choose the Reference treatment”:** the run is blocked until you choose the Reference treatment in the Bioequivalence settings. The app pre-selects it only for unambiguous names (R, Ref, Reference, Comparator, Innovator, Originator, RLD).

**No verdict:** see When There Is No Verdict, in Path 6. The Bioequivalent? column gives the reason.

**No verdict for my partial AUC:** the interval is marked supportive, or the metric has a value of zero in at least one profile. The table gives the reason and the number of zero values per treatment.

**“Treatment column must have exactly 2 levels”:** see Studies with Three or More Treatments.

## Figures

**Download Figure is grey:** it becomes active once the first plot has appeared; wait for it.

**Time points are missing from my summary plot:** the summary plot averages per time value. With actual sampling times (0.95 h and 1.05 h for a nominal 1 h), use a copy of the file with nominal times for the figure; keep the actual times for the NCA.

**“N observations with concentration  $\leq 0$  excluded”:** zeros, for example BLQ values set to 0 by Rules 1, 2 or 5, cannot be log-transformed, so they are left out of the geometric mean at that time point. They stay in your data.

**Too many colours to distinguish:** try the Colorblind-safe (Okabe-Ito) palette. With more than eight groups, switch to Grayscale and annotate the exported figure in a graphics program.

## Controlled Installation

**I cannot sign in:** check your user ID; after five wrong passwords the account is locked, and only the system owner can unlock it (reset).

**admin does not work for my new account:** someone may have used it first. Tell the system owner at once.

**I think someone knows my password:** change it at once (Change password in the menu with your name) and tell the system owner.

**I forgot my password:** ask the system owner for a reset, then sign in with admin and choose a new password.

**I was signed out while working:** nothing happened for 15 minutes. Upload the file again; records you created are stored on the server.

**"This action was not carried out because it could not be recorded in the audit trail":** nothing was done and nothing was lost. Tell the system owner.

**My record was rejected:** read the reason on the Records page, change the analysis, and create a new record.

**I signed the wrong record:** a signature cannot be undone. Tell the reviewer or the system owner; the analyst creates a new record, which is reviewed again. The mistaken signature stays visible in the audit trail.

**Signature INVALID:** the stored copy has changed since it was signed. Tell the system owner at once.

## Records and Exports

**The reproduction check says DIFFERENT:** a file in the record was changed after it was made, a parameter is present on one side only, or the results differ. `reproduction_check.txt` gives the reason.

**The record holds a CSV instead of my Excel file:** the uploaded file was no longer available; see Complete Analysis Record.

## Appendix E: Setting Up and Administering a Controlled Installation

For IT and the system owner. The installation checks of the IQ/OQ/PQ protocol (section 2.1) test the server against these steps, so follow them as written. Controlled mode is qualified per installation: a changed step is a deviation.

### Setting Up the Server

1. Install R, the system programs zip (the Analysis Records are built with it) and logger (security alerts go through it; on most Linux systems it comes with `util-linux` or `bsdutils`), and the validated package versions from `validation/renv.lock` into the site library, including `shinymanager`, `DBI` and `RSQLite`.
2. Put the tagged release in `/srv/shiny-server/nca`, owned by root and read-only for the service account shiny. Create the empty file `gxp/CONTROLLED` in it, owned by root: with it, the app does not start without controlled mode.
3. In `/etc/shiny-server/shiny-server.conf`, set `run_as` shiny, let Shiny Server listen on 127.0.0.1 only, and set `app_idle_timeout` 0 so that the app keeps running (otherwise every idle stop shows up as an exception in the audit trail).
4. Put nginx in front of it with HTTPS only, redirecting HTTP. Three settings differ from nginx's defaults: WebSocket forwarding (the app talks to the browser over a WebSocket), a read timeout of an hour so that an open session is not cut off, and an upload limit above the app's 50 MB (the default of 1 MB refuses most data files). The example below has all three.
5. Create the controlled directory on the server's local disk, never on a network or cloud-synced share: `mkdir /srv/nca-gxp`, owned by shiny, mode 0700.
6. Create `/home/shiny/.Renv`, mode 0600, with `NCA_GXP_DIR=/srv/nca-gxp`, `NCA_GXP_KEY` (a passphrase for the user store, kept nowhere else in plain text), `NCA_GXP_ORG` (your organisation's name, recorded in every audit entry) and, if needed, `NCA_GXP_WORK_HOURS` (for example 07:00-19:00).
7. Add a sudoers rule that lets the system owner run exactly `Rscript /srv/shiny-server/nca/gxp/manage_users.R` as shiny, and nothing else (example below). Root access stays with IT, with the operating system's audit logging.
8. Run `sudo -H -u shiny Rscript /srv/shiny-server/nca/gxp/manage_users.R init`. The `-H` matters: without it R reads the system owner's own `.Renv` and does not find the settings.
9. Turn on NTP time synchronisation, and a daily backup of `/srv/nca-gxp` to a separate site and network. Test a restore.
10. Make sure the server can only be reached through your two-factor sign-in, from managed devices whose browsers do not save passwords, and that the system log is monitored: security alerts are logged under the tag `nca-assistant`.

## Example Configuration

Adapt the names, paths and address to your server. Lines starting with # are comments.

/etc/shiny-server/shiny-server.conf:

```
run_as shiny;
server {
    listen 3838 127.0.0.1;          # only nginx on the same server can reach it
    location / {
        site_dir /srv/shiny-server;
        log_dir /var/log/shiny-server;
        app_idle_timeout 0;        # keep the app running between users
    }
}
```

/etc/nginx/conf.d/nca.conf (the app is then at <https://nca.example.org/nca/>):

```
map $http_upgrade $connection_upgrade { default upgrade; '' close; }

server {
    listen 80;
    server_name nca.example.org;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name nca.example.org;
    ssl_certificate      /etc/ssl/certs/nca.crt;
    ssl_certificate_key  /etc/ssl/private/nca.key;
    client_max_body_size 60m;      # the app accepts files up to 50 MB

    location / {
        proxy_pass http://127.0.0.1:3838/;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection $connection_upgrade;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 3600s;  # do not cut off an open session
        proxy_buffering off;
    }
}
```

/home/shiny/.Renviron (owner shiny, mode 0600):

```
NCA_GXP_DIR=/srv/nca-gxp
NCA_GXP_KEY=<a long passphrase, kept only here and in your password vault>
NCA_GXP_ORG=Your Organisation
NCA_GXP_WORK_HOURS=07:00-19:00
```

/etc/sudoers.d/nca (edit with visudo -f /etc/sudoers.d/nca; replace jdoe by the system owner's account, and /usr/bin/Rscript by the path that which Rscript shows):

```
jdoe ALL=(shiny) /usr/bin/Rscript /srv/shiny-server/nca/gxp/manage_users.R *
```

After a change to these files, restart Shiny Server and reload nginx. Then check that the app starts in controlled mode: the audit trail's first app\_started entry names the controlled directory, and a file of 5 MB or more uploads without an error.

## Managing Accounts

The system owner runs every command as: `sudo -H -u shiny Rscript /srv/shiny-server/nca/gxp/manage_users.R` command. Every change is logged under the system owner's own name, with the reason given. There is no delete: a user ID is never used again.

| Command                                           | What it does                                                                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>add id "Printed Name" roles "reason"</code> | Creates an account with the starting password admin. Roles: analyst, reviewer, analyst;reviewer or inspector. Create it only when the person can sign in at once |
| <code>role id roles "reason"</code>               | Changes the roles                                                                                                                                                |
| <code>reset id "reason"</code>                    | Sets the password back to admin and unlocks the account; the user must sign in at once and choose a new password                                                 |
| <code>deactivate id "reason"</code>               | Blocks sign-in and signing; the account and its history stay                                                                                                     |
| <code>list</code>                                 | Shows every account, its roles, status and history                                                                                                               |
| <code>verify</code>                               | Checks the audit trail; verify folder checks an archived copy, also on another computer                                                                          |
| <code>head</code>                                 | Prints the last entry of the trail: file it in your quality system at a fixed interval                                                                           |
| <code>archive folder</code>                       | Writes a read-only archive (below)                                                                                                                               |

Before you give out an account, check the person's identity. Deactivate accounts on the day someone leaves or no longer needs access, and review the account list at a fixed interval.

### Archiving and Restoring

`manage_users.R` archive folder writes a dated, read-only folder with a consistent copy of the audit trail, the stored records, the app files that are running, `validation/renv.lock`, the release manifest, the validation files, a verification report, restore steps (`RESTORE.md`) and a SHA-256 manifest of everything. Archive at the end of a study and when the server is retired, and keep the archive as long as the records. The restore steps bring the archived release back on a clean computer, on a copy of the archive; test this once during installation qualification.

### When Something Is Wrong

If `verify`, or `Verify` chain in the app, reports that the trail is not intact, stop using the installation for regulated work, keep the server as it is, and start an investigation under your procedure. Compare the trail with the heads you filed and with the backups to find out what changed. If the app says an action could not be recorded in the audit trail, check the disk space and the permissions of the controlled directory.

### A Training Installation

For training and for Tutorial 8, run controlled mode on your own computer. Use a separate copy of the app folder for it: the settings go in a file in that folder, so the app runs in controlled mode only when you start it from there, and your normal installation stays as it is. A training installation is not a qualified setup, because you own the folder that holds its audit trail.

On macOS and Linux:

1. Install the app as described in Installing the App on Your Own Computer. The installation script also installs shinymanager, DBI and RSQLite.
2. Copy the app folder, for example to ~/nca-training-app, and create an empty folder for the controlled data, for example ~/nca-training-data. Keep both on your local disk, not in OneDrive, Dropbox or a network folder: the audit trail can be damaged there.
3. In ~/nca-training-app, create a text file named .Renviron with these three lines (use the full path of your data folder):

```
NCA_GXP_DIR=/Users/yourname/nca-training-data
NCA_GXP_KEY=training-only-passphrase
NCA_GXP_ORG=Training
```

4. Open a terminal, go to the app folder and create the audit trail and two accounts:

```
cd ~/nca-training-app
Rscript gxp/manage_users.R init
Rscript gxp/manage_users.R add trainee1 "Trainee One" analyst "training"
Rscript gxp/manage_users.R add trainer1 "Trainer One" reviewer "training"
```

5. Start the app from the same folder: `Rscript -e "shiny::runApp(launch.browser = TRUE)"`. (In RStudio, open the folder as a project with File > New Project > Existing Directory: R then starts in this folder and reads its .Renviron file.)

6. Sign in as trainee1 with admin and choose a new password. Open a second browser, or a private window, and sign in there as trainer1. You can now do Tutorial 8.

On Windows the steps are the same, with these differences. Use forward slashes in the path in .Renviron (NCA\_GXP\_DIR=C:/Users/yourname/nca-training-data). Create .Renviron in the app folder with Notepad: choose All files as the file type, so it is not saved as .Renviron.txt. Run the commands in a Command Prompt in the app folder; if Rscript is not found, use its full path, for example "C:\Program Files\R\R-4.6.0\bin\Rscript.exe". Security alerts are written to security\_alerts.log in the data folder, because Windows has no logger.

After the training, delete both folders. They hold only training data; your normal installation is untouched.

## Appendix F: References

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## Appendix G: Document Control

Document ID NCA-UM-001. The PDF of this manual is part of each release of the app; its SHA-256 is listed in validation/release\_manifest.csv, so an installation can be checked to hold the manual of its release.

| Version | Date       | Author        | Changes                                                                                                                       |
|---------|------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | March 2026 | Rob ter Heine | Initial release                                                                                                               |
| 1.1     | April 2026 | Rob ter Heine | Added theoretical background (Part V), expanded statistical methods (Part VI), DDI study analysis, decimal separator guidance |
| 1.2     | April 2026 | Rob ter Heine | Quick Start data preparation callout, grid plot fix documentation, single-subject Analysis Record, DDI                        |

|     |                |               |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                |               | interpretation thresholds, regulatory DDI guideline links, troubleshooting updates                                                                                                                                                                                                                                                                                                                                                     |
| 1.3 | April 2026     | Rob ter Heine | New chapter 3 (Visualize Data); chapters 3–5 renumbered to 4–6; hub layout updated to 3+3 grid; attribution updated                                                                                                                                                                                                                                                                                                                    |
| 1.4 | April 2026     | Rob ter Heine | Factual corrections (Rule 6 label, AUC extrapolation flag); new feature documentation (steady-state in BE, balance check, design summary, Apply LLOQ button, excluded profiles note); beginner clarity additions (LLOQ/BLQ definitions, extravascular explained, R <sup>2</sup> failure consequences, SHA-256 explained, half-life review how-to, AUC % Extrapolated meaning)                                                          |
| 1.5 | September 2026 | Rob ter Heine | Updated for NCA Assistant v1.4.0: CDISC ADNCA upload and parameter codes, safety checks on upload, decimal comma and semicolon files, doses per period, replicate designs and the within-subject variability table, choosing the Reference treatment, the minimum R <sup>2</sup> rule, planning with Test and Reference CV, the reproduction check in the Analysis Record. Exercise 1 answers corrected; guideline references updated. |
| 1.6 | September 2026 | Rob ter Heine | Updated for NCA Assistant v1.5.0: partial AUCs in Paths 4, 5 and 6, with intervals ending at a time or at the last measurable concentration, C <sub>max</sub> and T <sub>max</sub> within an interval, pivotal and supportive metrics in bioequivalence, and interval shading in Visualize Data. New Exercise 4, new sections in Parts V and VI, and new glossary, troubleshooting and reference entries.                              |
| 1.7 | September 2026 | Rob ter Heine | Updated for NCA Assistant v1.6.0, after a review of the whole app. New: intended use and data notice; ICH M13A (C <sub>max</sub> and AUC to last point compared by default, BLQ values kept out of the half-life, checks listed with the results,                                                                                                                                                                                      |

|     |                   |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                   |               | no verdict where the guideline excludes one); widened limits for Cmax only; Rules 4 and 6 changed; IV bolus samples at time 0 set aside; thousands separators in decimal-comma files; unique subject IDs; units taken from the data; settings kept and results cleared; file integrity in the reproduction check; the Welch interval; excluding subjects. Exercise answers checked against v1.6.0.                                                                                                                                                    |
| 1.8 | 25 September 2026 | Rob ter Heine | For NCA Assistant v1.7.0, after a review by a tutor, a clinical pharmacologist and a regulatory compliance officer. New: Getting Started with local installation, seven tutorials with an answer key, study types (IV bolus and infusion, single and multiple ascending dose, food effect, metabolites, bioavailability, three or more treatments), Reading Your Results, a chapter on regulated use, derived-parameter formulas, an appendix of output parameters, and troubleshooting by stage. Corrected throughout for the app changes of v1.7.0. |
| 1.9 | September 2026    | Rob ter Heine | For NCA Assistant v1.8.0. New: chapter Working on a Controlled Installation (sign-in, records for review, review signatures, signature sheets, the audit trail page), Tutorial 8, Appendix E on setting up and administering a controlled installation, troubleshooting for controlled installations. The chapter on regulated use describes the controls of a controlled installation and the organisation's responsibilities.                                                                                                                       |